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Chapter 8

GELFAND-P000554

2/IX-74. Gelfand's sem. (32nd year of the seminar)

Giusti.

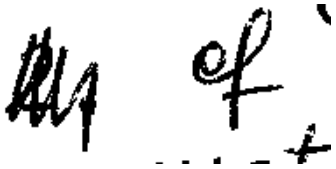
1. Singularities [crossed out] and regularity



Uncertain reading. Crossed-out strokes lie between "Singularities" and "and". The original entry is illegible.

Original source details: GELFAND-P000554

[crossed out] of solutions of non-linear elliptic (and parabolic) systems.



Uncertain reading. A separate crossed-out entry precedes "of solutions". Its reading remains unresolved.

Original source details: GELFAND-P000554

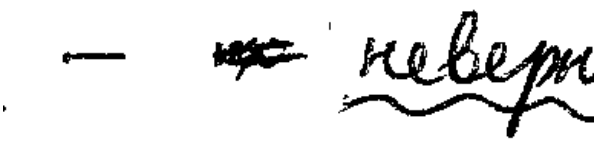
Nash, De giorgi, Moser. — Regularity: 1 equation linear ~~di...~~ with discont. coeff. (Höld. coeff.) ⇒ regul. for non linear with smooth coeff.



Uncertain reading. There are crossed-out letters before "with". The opening provisionally reads "di".

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Systems — [crossed out] false. There are [?] coeff. and [?] solutions



Uncertain reading. A heavily crossed-out sign follows the dash before "false".

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Original source details: GELFAND-P000554

Uncertain reading. The Russian abbreviations before "coefficients" and "solutions" provisionally read "Быв. ср." and "несгр.".

1968 Miranda, Giusti — syst. with analytic coeff. with discont. solutions



Uncertain reading. The beginning of the surname Giusti is overwritten and the underlying strokes are illegible.

Original source details: GELFAND-P000554

Σ — singular set.

(1) Σ closed
(2) $\dim \Sigma < n - 2$

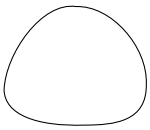
2. Boundary value problems. for surfaces of prescribed mean curvature.

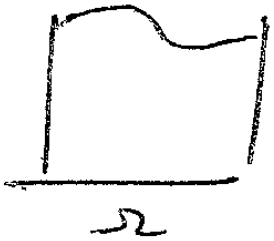
$u : \Omega \longrightarrow \mathbb{R}$ graph $u = S$



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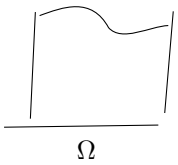
Diagram and typed description.





Original source details: GELFAND-P000554

Diagram and typed description.



$$H(x) = \frac{1}{n} \sum \frac{\partial}{\partial x_i} \left(\frac{\partial u / \partial x_i}{\sqrt{1 + |\operatorname{grad} u|^2}} \right)$$

H is not very large (otherwise unsolvable). Depends on the geometry of the domain, the func. $H(x)$ and so on

That is what will come next time.

S. P. Novikov. (and Dubrovin).

Abelian var. $\xi_1, \ldots, \xi_{2n}$ — basis in $\mathbb{C}^n$ (over $\mathbb{R}$), T^{2n} — corresp. torus. Let the Abelian var. (algebraic var.).

How to describe all tori?

$n = 2$ already nontrivial.

$y^2 = P_{2n+1}(x)$ Riem. surface $\Rightarrow$ torus.



Undeciphered passage. A crossed-out entry before H^1 provisionally begins “Вид...”. The rest is illegible.

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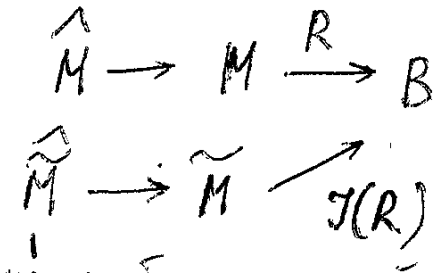
$$\underbrace{H^1(R,\mathbb{C})/H^1(R,\mathbb{Z})}_{\text{the torus is like this}} \qquad R \longrightarrow J^{2n}(R). \quad \text{(torus)}$$

$$k = 0, \dots, n-1, \qquad \boxed{\omega_k = \frac{x^k\,dx}{\sqrt{P_{2n+1}(x)}}} \quad \longleftarrow \quad \text{everywhere regular differentials.}$$

Basis — the forms ω_k.

$$P_{2n+1}(x) = \prod (x - x_\alpha).$$

Everything is det. by the set of roots.



Original source details: GELFAND-P000554

Diagram and typed description.

$$\begin{array}{ccccc} \widehat{M} & \longrightarrow & M & \xrightarrow{R} & B \\ & & & \nearrow & \\ \widehat{\widetilde{M}} & \longrightarrow & \widetilde{M} & & J(R) \\ | & & & & \end{array}$$

The lower diagonal arrow ends below and to the left of the upper B. Its exact connection remains unresolved.

M — collection of Riem. surfaces

B — Siegel upper half-space

(cover of the moduli set).

from the set of roots one constructs the polynomial, and from it the torus.

GELFAND-P000555

$$p_1, p_2, q_1, q_2, c, d. \qquad \mathbb{C}^6 \text{ or } \mathbb{C}^4_{c,d}$$

$$J_1 = p_1 p_2 + V(q_1, q_2)$$

$$J_2 = p_1^2 - 2q_1 p_1 p_2 + 2(q_2 - c)p_2^2 + \mathcal{D}(q_1, q_2)$$

$$\Omega = \sum_{\alpha=1}^{2} dp_\alpha \wedge dq_\alpha.$$

$\langle J_1, J_2 \rangle \equiv 0.$ Poisson br.

$$V = -\left(\frac{q_2^2}{2} + \frac{5}{2}q_1^2 q_2 + \frac{5}{2}q_1^4\right) + c_1 q_1^2 - d_1 q_1$$



Original source details: GELFAND-P000555

Uncertain reading. The sign before the equals sign is overwritten. It has two upper strokes and provisionally reads V.

$$\mathcal{D} = q_1^5 + 2c_1 q_1^3 + d_1 q_1^2 - 4q_1 q_2^2 + 4c_1 q_1 q_2 - 2d_1 q_2$$

$[X_{J_1}, X_{J_2}] = 0.$ Obtain 2 commut. fields.

The corresp. orbits — an Abelian var.

$$\left.\begin{array}{l} J_1 = \text{const} \\ J_2 = \text{const} \end{array}\right\} \quad \text{torus.}$$



Uncertain reading. A crossed-out entry left of the equations starts with a J-like form. The rest is illegible.

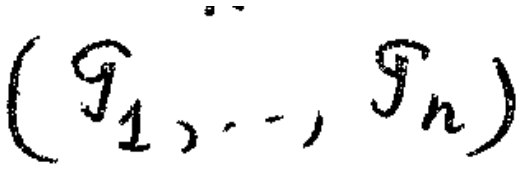
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Corollary: the coll. of them all — a unirat. variety.

$$R: \quad y^2 = x^5 + \frac{c_1}{4}x^3 - \frac{d_1}{16}x^2 + \left(\frac{J_1}{32} + \frac{c_1^2}{4}\right)x + \frac{J_2}{2^8} + \frac{c_1 d_1}{2^7}$$

R — Riem. surface, $S^n R$ — symm. power of R.

$(J_1, \ldots, J_n)$ and symmetrization.



Uncertain reading. Both letters in the row before “and symmetrization” have a hook. The reading J is provisional.

Original source details: GELFAND-P000555

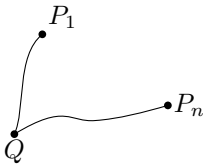
$$S^n R \longrightarrow J(R).$$

$(P_1, \ldots, P_n) \xrightarrow{A} (\alpha_1, \ldots, \alpha_n)$ Abel map.

(Abel’s theor. — it is an isomorphism).



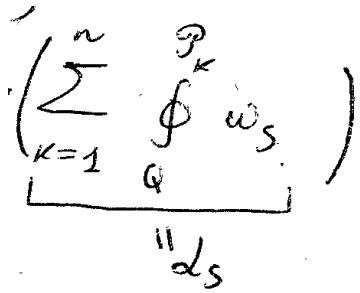
Diagram and typed description.



The upper curve thickens at P_1. An arrowhead cannot be distinguished.

Original source details: GELFAND-P000555

$$(P_1, \dots, P_n) \longrightarrow \left(\underbrace{\sum_{k=1}^{n} \oint_Q^{P_k} \omega_s}_{\| \ \alpha_s} \right)$$



Manuscript excerpt. A loop is visible on the integral sign. Two short strokes stand above α_s below the bracket.

Original source details: GELFAND-P000555

$$L = -\frac{d^2}{dx^2} + u(x),$$

$u(x+T) = u(x)$ or quasiperiodic.

Eigenfunc. $L\Psi(x, E) = E\Psi,$

$$\Psi(x+T) = \lambda\Psi(x), \qquad |\lambda| = 1, \qquad \lambda = e^{ip(E)}$$

(Bloch eigenfuncs.)

For almost periodic [crossed out] the collection of such eigenfuncs. does not form a complete system.



Undeciphered passage. A short densely crossed-out entry follows the Russian abbreviation for “periodic”.

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$\Psi|_{x_0} = 1$ — normalization

$\Psi(x, E, x_0)$ $(\Psi, \overline{\Psi})$ — 2 eigenfuncs.

branches in E at the endpoints of the spectral intervals. Obtain a Riem. surface of inf. genus. 2 sheets, since there are 2 eigenfuncs.

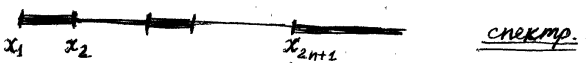
GELFAND-P000556

The set of pot. with given band structure — the Jacobian var. of this Riem. surface (there are formulas)



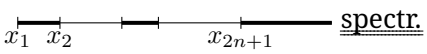
Uncertain reading. The first strokes in the Russian word for “band” are overwritten.

Original source details: GELFAND-P000556



Original source details: GELFAND-P000556

Diagram and typed description.



Pot. with given band structure — the corresp. torus (Jacobian var.). .....

9/IX-74 . Gelfand's sem..

Bessaga (Warsaw). Anderson's Z-sets and their applications. (R. D. Anderson)

X — a complete metriz. space. $A \subset X$, $\overline{A} = A$.

A is called a Z-set if $\forall$ cont. maps

$$f : \underbrace{I^\infty}_{\text{Hilb. cube}} \longrightarrow X, \qquad \forall \varepsilon > 0, \qquad \exists g : I^\infty \longrightarrow X \setminus A, \qquad \rho(f,g) < \varepsilon.$$

$$Q = [-1,1]^\infty, \qquad P = (-1,1)^\infty. \qquad (Q \setminus P) \text{ — } Z\text{-set}$$

Prop.: 1°. If A, B are Z-sets (in Q), then $f : A \xrightarrow{\sim} B$ extends to a ~~homeo~~ homeom. of the entire space.



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Uncertain reading. The numeral of 1° is heavily overwritten. The small circle and period are distinguishable.



Original source details: GELFAND-P000556

Uncertain reading. The crossed-out abbreviation before "homeom." provisionally reads "homeo".

Z_σ-set — a countable union of Z-sets.

2°. There exists a maximal Z_σ-set _____ · _____

Gelfand

Consid. $\mathbb{R}^\infty$ in the Tych. topology and the space S of rays [crossed out] issuing from 0 in the nat. topology. Prove that on S every cont. func. is [crossed out] const



Original source details: GELFAND-P000556

Undeciphered passage. There is a densely crossed-out entry between "rays" and "issuing".



Original source details: GELFAND-P000556

Uncertain reading. A small crossed-out mark precedes "const". Its reading remains unresolved.

(Gelf., Fuks. Funct. Anal. I, 4). _____ · _____

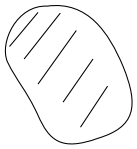
Bessaga. All separab. Fréchet spaces are homeomorphic.

Giusti. $\Omega \subset \mathbb{R}^n$



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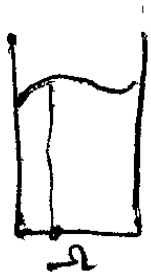
Diagram and typed description.



$$\begin{cases} \sum_{i=1}^{n} D_i \left\{ \dfrac{D_i u}{\sqrt{1+|Du|^2}} \right\} = -H(x,u) \quad \text{in } \Omega, \\ u = \phi \quad \text{on } \partial\Omega \end{cases} \qquad D_i = \frac{\partial}{\partial x_i}.$$

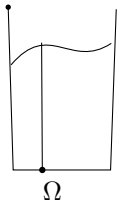
n times the mean curvature $=$ the left-hand side of the eq.

$H \equiv 0 \quad \Rightarrow \quad$ minimal surface.



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Diagram and typed description.



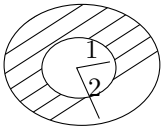
A dark mark sits at the lower end of the inner vertical stroke. Its shape is indistinct.

Example 1. $\Omega, \quad n = 2, \quad H = 0$



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Diagram and typed description.

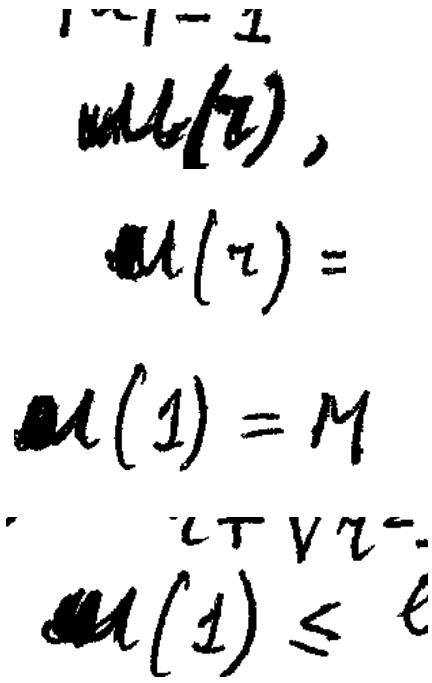


circular annulus. The labels 1 and 2 stand beside the two radial strokes.

$$\phi = \begin{cases} 0 & |x| = 2 \\ M & |x| = 1 \end{cases}$$

Solution: $u(r)$, $r = |x|$.

$$u(r) = c \log \frac{2 + \sqrt{4 - c^2}}{r + \sqrt{r^2 - c^2}}, \qquad 0 \le c \le 1,$$
$$u(1) = M \quad \Rightarrow \quad u(1) \le \log(2 + \sqrt{3}) = M_0.$$



Uncertain reading. The opening strokes of $u(r)$ and $u(1)$ are heavily overwritten. The underlying forms are illegible.

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GELFAND-P000557

For $M > M_0$ there are no solutions.

Exper. check: soap bubble

we carefully lift the stick steak.



Manuscript excerpt. Below the Russian word for “stick” the manuscript reads “steak.” with a wavy underline.

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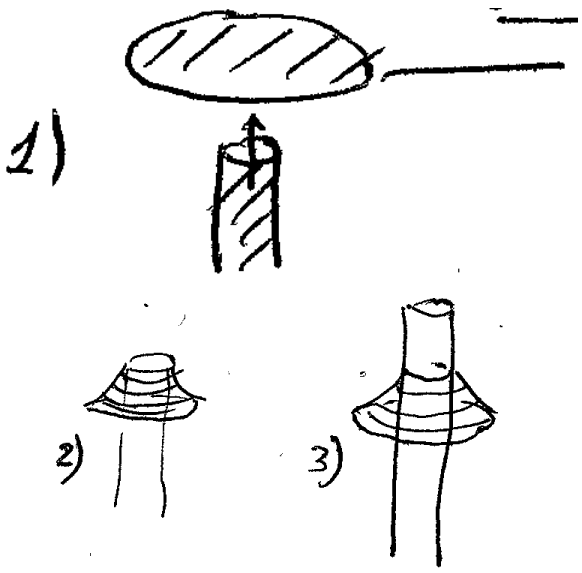
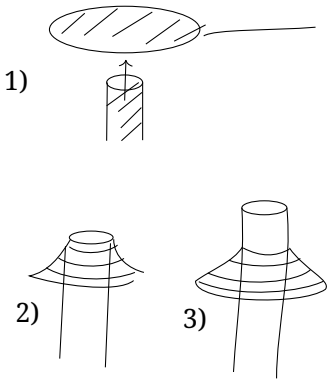


Diagram and typed description.

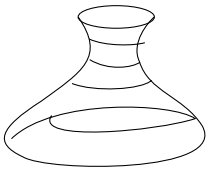


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if it is a wire ring, then



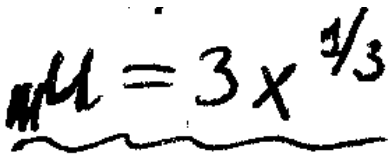
Diagram and typed description.



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Example 2. $n = 1, \quad \Omega = (-1, 1)$

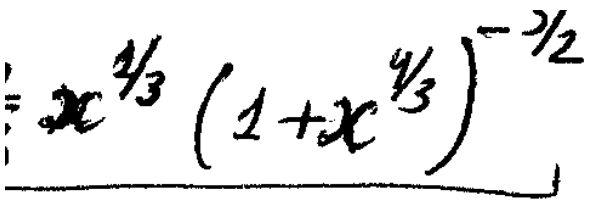
$u = 3x^{1/3}$



Uncertain reading. The initial letter is overwritten. Its final outline provisionally reads u.

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$$\left(\frac{u'}{\sqrt{1+u'^2}}\right)' = \underbrace{-\frac{2}{3}x^{1/3}\left(1+x^{4/3}\right)^{-3/2}}_{\text{cont.}}$$



Manuscript excerpt. The exponent on the first x is $1/3$. The exponent on the second x is $4/3$.

Original source details: GELFAND-P000557

[crossed out] u — nondiff.



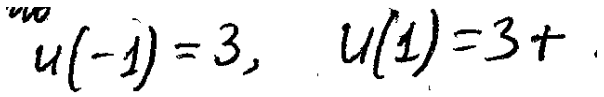
Uncertain reading. The crossed-out Russian entry may begin “об...”. Its end is illegible.

Original source details: GELFAND-P000557

$$u(-1) = -3, \qquad u(1) = 3$$

Let

$$u(-1) = 3, \qquad u(1) = 3 + A$$



Manuscript excerpt. After “Let” the manuscript reads $u(-1) = 3$.

Original source details: GELFAND-P000557

$$\frac{u'}{\sqrt{1+u'^2}} = (1+x^{4/3})^{-1/2}$$

$$u' \le x^{-2/3}$$

$$u(x) \le -3 + \int_{-1}^{x} t^{-2/3}\,dt = 3x^{1/3}$$

_____ · _____

$$\mathcal{F}(u) = \int_\Omega \sqrt{1+|Du|^2}\,dx + \int_\Omega \lambda(x,u)\,dx$$

Need to find min.

$$\lambda(x,t) = \int_0^t H(x,s)\,ds$$

Under the cond. $|u'| \le K$ there is a min. (otherwise none).

u_k — soln. Let $k \to +\infty$. $\quad u_k \to u_0$? _____ · _____

u — Lip-contin. solut. $\quad A \subset \Omega$

$$\int_A H(x)\,dx = \int_{\partial A} n_i \frac{D_i u}{\sqrt{1+|Du|^2}}\,dH_{n-1}$$

$$\left|\int_A H(x)\,dx\right| \le (1-\varepsilon_0) H_{n-1}(\partial A)$$

$$\boxed{1-\varepsilon_0 = \sup_\Omega \frac{|Du|}{\sqrt{1+|Du|^2}}.}$$

Therefore H is not very large. _____ · _____

GELFAND-P000558

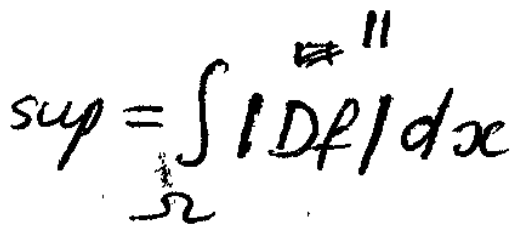
① $H^{1,1} = W_1^1$ ‖

$$\mathcal{F}(u) = \int_\Omega \sqrt{1+|Du|^2}\,dx + \int_\Omega \lambda(x,u)\,dx + \int_{\partial\Omega} |u-\phi|\,dH_{n-1}$$

$BV(\Omega) = L^1$ funcs., whose first derivs. are measures (Radon).

$$\text{i.e.} \quad f \in BV \iff f \in L^1,$$
$$\sup\left\{\int_\Omega f \operatorname{div} \vec{g}\,dx, \quad \vec{g} \in C_0^1(\Omega),\ |g| \le 1\right\} < +\infty$$

$$\sup = \int_\Omega |Df|\,dx$$

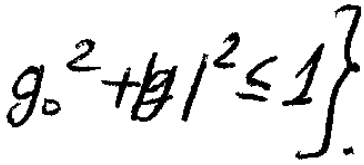


Uncertain reading. Two separate strokes stand above Df. The crossed-out mark to their left is illegible.

Original source details: GELFAND-P000558

Area $f \in BV$.

$$\mathcal{A}(f) = \int \sqrt{1+|Df|^2}\,dx \underset{\text{def}}{=}$$
$$\sup\left\{\int_\Omega (g_0 + f \operatorname{div} \vec{g})\,dx, \quad g_0, g \in C_0^1, \quad g_0^2 + |g|^2 \le 1\right\}.$$



Uncertain reading. The left stroke at g merges with the letter. The reading $|g|^2$ is provisional.

Original source details: GELFAND-P000558

area.

$(\mathcal{A}f)$ lower semicont. with respect to L^1 ~~cont...y~~.



Uncertain reading. The crossed-out English word provisionally reads “cont...y”. The full word remains unresolved.

Original source details: GELFAND-P000558

② Bounded sets in BV are relatively compact in L_1

. . . . . .

Dubrovin.

$$L = -\frac{d^2}{dx^2} + u(x), \qquad u(x+T) = u(x)$$

E — spectral parameter.

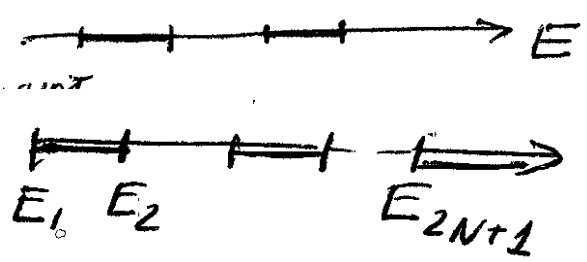
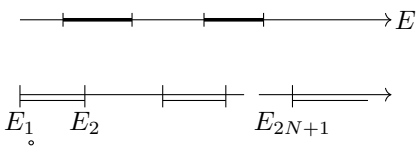


Diagram and typed description.



A small unfilled loop stands below E_1. Its meaning remains unresolved.

Original source details: GELFAND-P000558

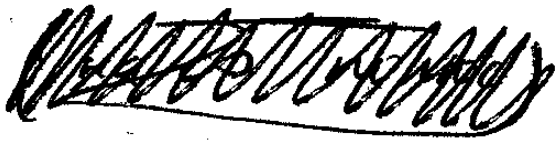
Sometimes the gaps disappear.

Let $u(x)$ — a finite-gap potential

Let $R(E) = \prod(E - E_i)$

Consid. the Riem. surface of the func. $\sqrt{R(E)}$ — denote by Γ_N.

Bloch eigenfuncs. continue analyt. to $\Gamma_N \setminus \infty$ [crossed out]



Undeciphered passage. The parenthetical entry at the right is densely crossed out and illegible.

Original source details: GELFAND-P000558

(meromorphically)

N — genus.

and has N poles and N zeros

Bloch eigenfunc.: $\Psi(x, E, x_0)$

Poles $\gamma_i(x_0)$. || there is a natural proj.

$$\Gamma_N \longrightarrow \mathbb{C}_E.$$

2-sheeted covering

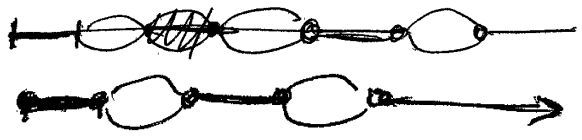
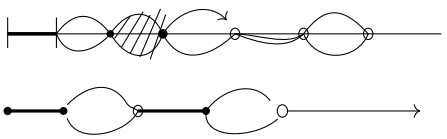


Diagram and typed description.



Original source details: GELFAND-P000558

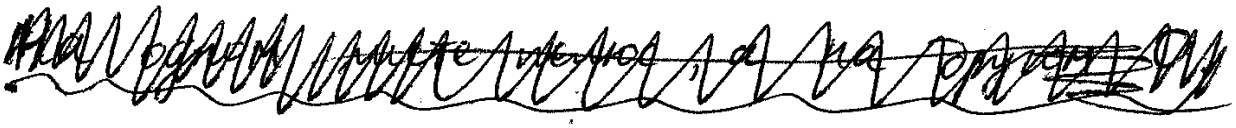
An oval projects to a gap: on each of them lies one pole. The loc. of the poles depends on the normalization: x_0.

~~On one sheet [...], and on the other [...]~~



Original source details: GELFAND-P000558

Uncertain reading. The crossed-out line provisionally begins “On one sheet”.



Original source details: GELFAND-P000558

Undeciphered passage. There are isolated strokes after “sheet”. The next legible words provisionally read “and on the other”. The full reading remains unresolved.

On each oval also one zero. _____ · _____

Next time 18^{30}

GELFAND-P000559

16.9.74. Gelfand Sem. Dubrovin.

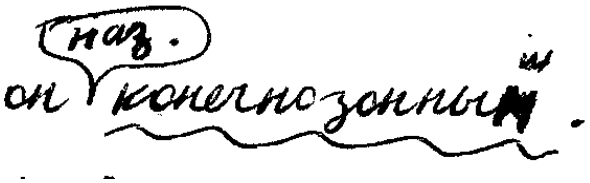


Original source details: GELFAND-P000559

Uncertain reading. The second digit of the date has been written over with a heavy loop. The current reading “6” is provisional. The earlier mark is illegible.

$$\text{Let} \quad L = -\frac{d^2}{dx^2} + u(x), \qquad u(x+T) = u(x).$$

The spectrum L — a set of intervals. If there are finitely many of them, it is [inserted above: called] finite-gap.



Original source details: GELFAND-P000559

Uncertain reading. A letter at the end of the Russian word is heavily crossed out. The insertion above provisionally reads “й”. The earlier ending is illegible.

The spectrum — those λ for which there is a bounded soln. of the eq. $Ly = \lambda y$.

$$y_1(x+T) = \alpha_{11} y_1(x) + \alpha_{12} y_2(x)$$
$$y_2(x+T) = \alpha_{21} y_1(x) + \alpha_{22} y_2(x).$$

The matrix $(\alpha_{ij}) = \widehat{T}$ (of monodromy).

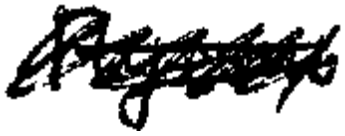
We need $\widehat{T}^n y$ to be bounded, i.e. $\exists$ an eigenvalue with $|\lambda| = 1$

but: $\det \widehat{T} = 1$ (Wronskian); $\lambda_1 \lambda_2 = 1$.

$$\lambda^2 - (Sp\,\widehat{T})\lambda + 1 = 0$$

GELFAND-P000560

~~Roots~~



Original source details: GELFAND-P000560

Uncertain reading. The crossed-out word provisionally reads “Roots”. Crossing strokes obscure some letters.

$$|\alpha_{11} + \alpha_{22}| < 2 \quad \Rightarrow \quad \text{complex roots.}$$
$$|\alpha_{11} + \alpha_{22}| > 2 \quad \Rightarrow \quad \text{real roots.}$$

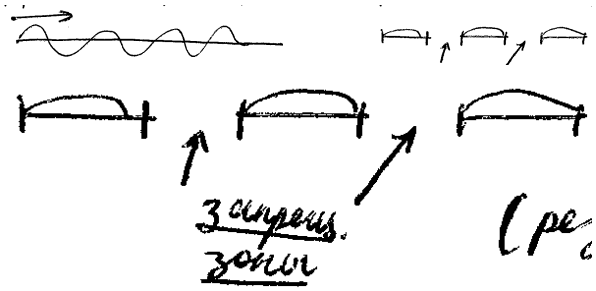
The roots run in one direction (can be proved).

Phys. interpret.: if there is a bounded eigenfunc., the particle will pass through, if not, it will ~~not~~ get trapped



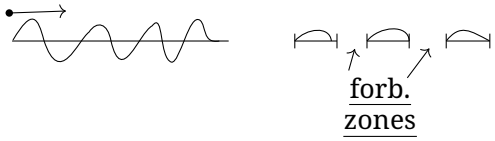
Original source details: GELFAND-P000560

Uncertain reading. The short crossed-out word before the boxed word provisionally reads “not”.



Original source details: GELFAND-P000560

Diagram and typed description.



exp decrease forb. zones. (resonance of particle and field)



Original source details: GELFAND-P000560

Uncertain reading. At the start of the Russian word for “decrease”, there is a high stroke above overwritten marks. The earlier spelling is illegible.

Ince, Akhiezer, Hochstadt.

$$\text{Bloch func.} - \boxed{u(x+T) = \nu u(x)}, \qquad |\nu| = 1. \quad (\text{eigenvector } \widehat{T}).$$

$$\nu = e^{ip}, \qquad p \text{ — “quasimomentum”.}$$

Bounded eigenfuncs. — if $p \in \mathbb{R}$.

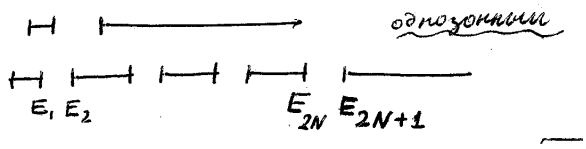
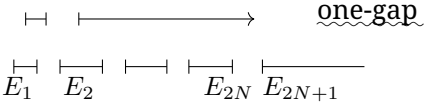


Diagram and typed description.



Original source details: GELFAND-P000560

$$\Gamma_N \text{ — Riemann surface of } \sqrt{R(E)}, \qquad \text{where} \quad R(E) = \prod_1^{2N+1} (E - E_i)$$

G. Let $y(x, E)$ — Bloch func. for complex E.

$$y(x+T) = \nu y(x)$$

$y(x_0, x, E)$ normalization: $y(x_0, x_0, E) = 1$.

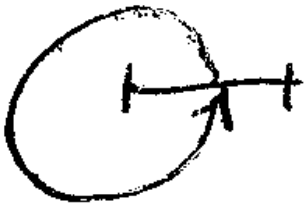
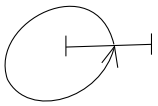


Diagram and typed description.



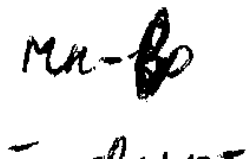
Two short diverging strokes are visible where the loop meets the segment. The mark provisionally points upward.

Original source details: GELFAND-P000560

D. Th. 1. Bloch eigenfuncs. continue analyt. to Γ_N.

Th. 0. Any finite-gap pot. — a stat. soln. of a higher KdV eq..

Th. 2 The set of pots. with a given spectrum is $J(\Gamma_N)$ — the Jacobi manifold.



Uncertain reading. In the Russian abbreviation for “set”, the letter after the hyphen was written over. The earlier strokes are illegible.

Original source details: GELFAND-P000560

KdV (dyn. syst.) and its higher analogues act there.
On solns. $Ly = Ey$ acts $(\widehat{T}y)(x) = y(x+T)$

$$\begin{matrix} c=1, & c'=0 \\ s=0, & s'=1 \end{matrix} \quad \text{(at pt. } x_0) \qquad c(x, x_0, E)$$

In this basis the monodromy matr. $\alpha_{ij}(E, x_0)$.
Everything is det. by the matr. trace $\tau = \frac{1}{2} Sp\, \alpha_{ij}$

$$|\tau| \leq 1 \quad \Rightarrow \quad \text{spectral bands}$$



Uncertain reading. The ending of the Russian word for "bands" is overwritten. The earlier strokes are illegible.

Original source details: GELFAND-P000560

τ is indep. of x_0, $\qquad \tau = \tau(E)$.

$$\lambda^2 - 2\tau\lambda + 1 = 0$$

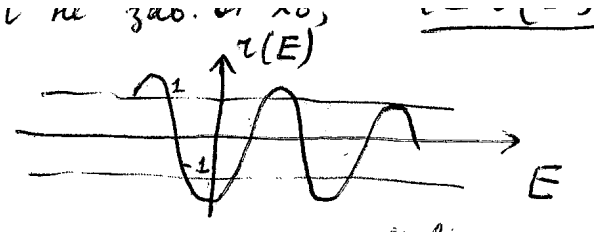
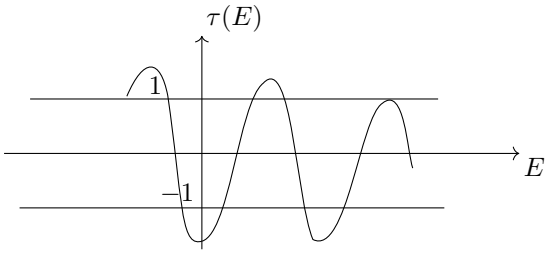


Diagram and typed description.



Original source details: GELFAND-P000560

it is known that the graph looks like this — max and $min \geqslant 1$ — prob.



Uncertain reading. The sign before 1 consists of an upper stroke and two oblique strokes. The provisional reading is $\geqslant$.

Original source details: GELFAND-P000560

GELFAND-P000561

for the 2^* year There are always infinitely many of them (max and min).



Uncertain reading. A small crossed mark follows "2". The visible shape is retained. The ending remains unresolved.

Original source details: GELFAND-P000561

merger of 2 forb. zones — there is no allowed zone

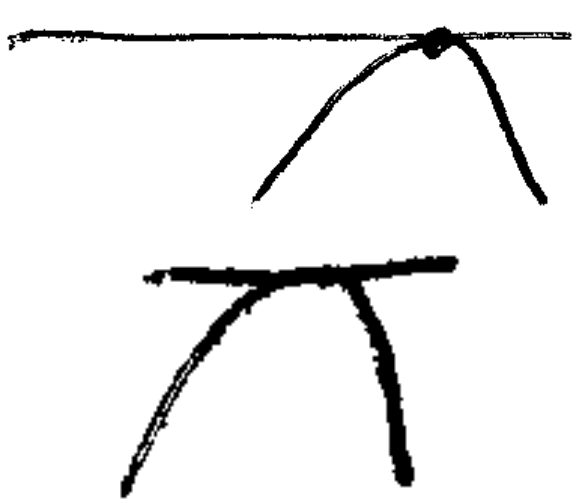
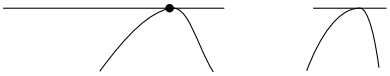


Diagram and typed description.



Original source details: GELFAND-P000561

From the eq. $\tau(E) = \pm 1$ one must select the simple roots to det. the band edges.
The band edges give a Jordan block. If diagonal, we get the degenerate case.
See the work of Gelfand and Lidskii (1955) on systems with periodic coeffs..
Def. $\Psi(x, x_0, E), \qquad \Psi(x+T) = e^{ip}\Psi(x), \qquad \Psi(x_0) = 1.$
The other is complex conjugate (when $|\tau(E)| < 1$).

$$\text{Let} \qquad \chi = -i\frac{\Psi'}{\Psi}$$

1. $\chi = \chi(x, E)$ (indep. of x_0).

Uncertain reading. A detached ", and" stands to the right between the first and second items. Its attachment remains unclear.



Original source details: GELFAND-P000561

2. $-i\chi' + \chi^2 + u - E = 0$ (Riccati eq.).

3. $\chi_I = \dfrac{1}{2}\dfrac{\chi_R'}{\chi_R}, \qquad \chi = \chi_R + i\chi_I.$

4. $\chi(x, E) \sim k + \sum_{n=1}^{\infty} \dfrac{\chi_n(x)}{(2k)^n}, \qquad$ where $k = \sqrt{E}$

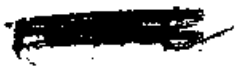
Uncertain reading. A wavy stroke lies over earlier strokes before k. The reading $\sim$ is provisional.



Original source details: GELFAND-P000561

5. $\chi(x+T)=\chi(x)$. (asympt. expansion).

$$2 \Rightarrow 3, \qquad 2 \Rightarrow 4$$



Undeciphered passage. A group of signs after $2 \Rightarrow 4$ is heavily crossed out. The surviving strokes are insufficient to read it.

Original source details: GELFAND-P000561



Uncertain reading. A short group is crossed out before χ_R. It provisionally reads χ with a subscript, possibly R.

Original source details: GELFAND-P000561

$$\chi_R(x,E) = k + \sum_{n=0}^{\infty} \frac{\chi_{2n+1}(x)}{(2k)^{2n+1}}$$

$$\Psi = \exp\left(i\int_{x_0}^{x} \chi\, dx\right) = \sqrt{\frac{\chi_R(x_0)}{\chi_R(x)}}\exp\left(i\int_{x_0}^{x} \chi_R\, dx\right).$$

1. $|\Psi|^2 = \dfrac{\chi_R(x_0)}{\chi_R(x)}$

2. $p(E) = \displaystyle\int_0^T \chi_R(x,E)\, dx + \pi n$

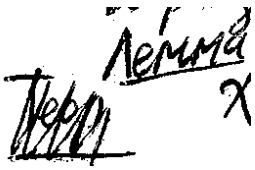


Uncertain reading. A heavy slanted mark precedes πn. A separate factor remains unresolved.

Original source details: GELFAND-P000561

Express χ in terms of the monodromy matr.

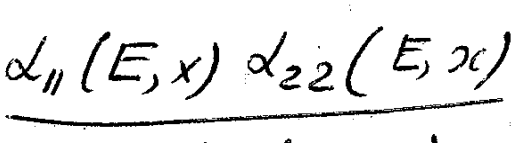
~~Th.~~ Lemma



Uncertain reading. The underlined “Th.” is crossed out. “Lemma” is inserted above and to its right.

Original source details: GELFAND-P000561

$$\chi(x,E) = \frac{\sqrt{1-\tau^2(E)}}{\alpha_{21}(E,x)} + \frac{i}{2}\frac{\alpha_{11}(E,x)\alpha_{22}(E,x)}{\alpha_{21}(E,x)} \tag{1}$$



Uncertain reading. A separate sign is indistinguishable between $\alpha_{11}(E,x)$ and $\alpha_{22}(E,x)$. The groups are typeset side by side.

Original source details: GELFAND-P000561

Proof. $\Psi(x, x_0, E)$ — an eigenvector of the monodr. matr. Use the rules of lin. algebra:

$$\Psi(x, x_0, E) = c(x, x_0, E) + i \begin{pmatrix} \text{right-hand side of} \\ \text{formula (1) at pt. } x_0 \end{pmatrix} s(x, x_0, E).$$

GELFAND-P000562

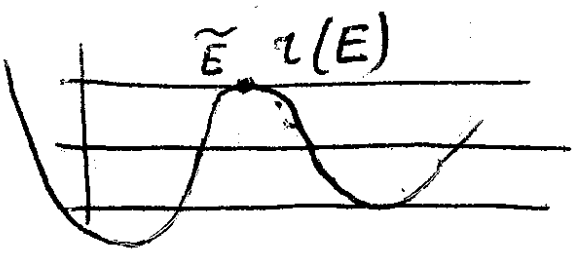
Find the Wronskian det. of this func. and c (it is indep. of x).

$$\{c, \Psi(x, x_0, E)\}|_{x_0} = \quad \text{everything works out.}$$
$$i\chi(x_0, E) = i(\ldots)$$
$$\{c, \Psi\}|_{x_0} = i\Psi'(x_0) = i\chi(x_0).$$



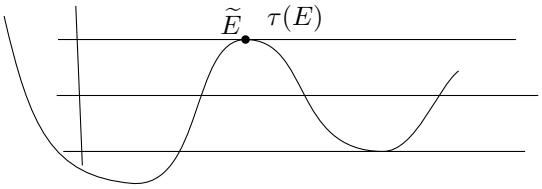
Original source details: GELFAND-P000562

Uncertain reading. An overwritten mark with a prime follows i. The current reading Ψ' is provisional.



Original source details: GELFAND-P000562

Diagram and typed description.



If $(1 - \tau(E))$ has a multiple zero, then the monodromy matr. is diagonal, i.e.

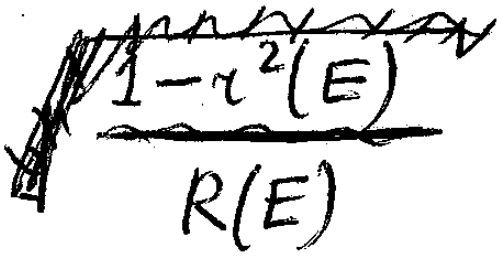


Original source details: GELFAND-P000562

Uncertain reading. A short mark resembling "π" is crossed out before the equalities. Its reading remains unresolved.

$$\alpha_{12}(\widetilde{E}, x_0) = 0, \qquad \alpha_{21}(\widetilde{E}, x_0) = 0.$$

$\dfrac{1 - \tau^2(E)}{R(E)}$ has only double degen. zeros.



Original source details: GELFAND-P000562

Uncertain reading. An oblique stroke and an upper line are heavily crossed out to the left of and above the fraction. They may be a deleted radical sign.

$\sqrt{\dfrac{1-\tau^2(E)}{R(E)}}$ has only simple zeros. (see Titchmarsh)

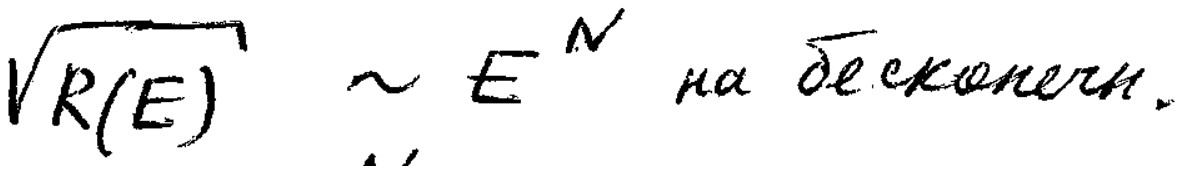
$\alpha_{21}(\widetilde{E}, x_0) = 0$ too

$$\alpha_{21} = \widetilde{\alpha}_{21} \cdot \sqrt{\frac{1-\tau^2(E)}{R(E)}} \qquad \text{(divided)}.$$

Substitute into the formula for χ_R:

$$\chi_R = \frac{\sqrt{R(E)}}{\widetilde{\alpha}_{21}(E,x)}, \qquad \text{where } \widetilde{\alpha}_{21}(E,x) \text{ — an entire anal. func. of } E.$$

$$\chi_R = k + O\left(\frac{1}{k}\right), \qquad \sqrt{R(E)} \sim E^N \quad \text{at infinity.}$$



Original source details: GELFAND-P000562

Uncertain reading. The visible exponent in this line is N. The numerator in the next line is kE^N.

$$\chi_R(x,E) \sim \frac{kE^N}{E^N} \quad \Rightarrow \quad \widetilde{\alpha}_{21}(E,x) \text{ — a polynomial in } E.$$

$P(E,x)$

$\chi_R(x,E)$ — 2-valued analyt. func. (analyt. continuation from real E in a band)

$$\chi_R(x,E) = \frac{\sqrt{R(E)}}{P(E,x)}$$

$$P(E,x) = \prod (E - \gamma_i(x)), \qquad \gamma_i(x) \text{ — roots of the polynomial}$$

Hence : $\boxed{\chi(x,E) = \left(\sqrt{R} - \frac{i}{2}\frac{d}{dx}P\right)/P.}$

$$\chi_R \sim k + \sum \frac{\chi_{2n+1}(x)}{(2k)^{2n+1}}$$



Uncertain reading. The letter n in the subscript $2n+1$ has been written over. The earlier strokes are illegible.

Original source details: GELFAND-P000562

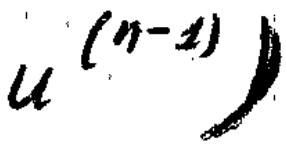
Construct the functionals

$$I_n\{u\} = \int_0^T \chi_{2n+3}(x)\,dx$$

GELFAND-P000563

$$\underbrace{-i\chi' + \chi^2 + u - E = 0} \quad \Rightarrow \quad \text{asymptotics.}$$

and in this case $\chi_{2n+1} = Q(u, u', \ldots, u^{(n-1)})$ — a polynomial.



Original source details: GELFAND-P000563

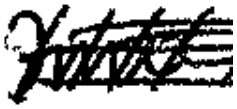
Uncertain reading. The order above the last u provisionally reads $(n-1)$.



Original source details: GELFAND-P000563

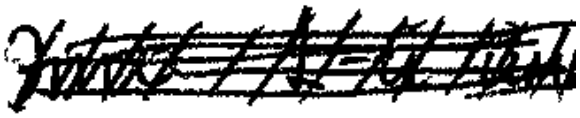
Uncertain reading. A group with vertical strokes and a loop is crossed out at the left. The visible sign resembles "δ" or "$\int$". The earlier writing remains unresolved.

~~This [...]~~



Original source details: GELFAND-P000563

Uncertain reading. The beginning of the crossed-out line provisionally reads "This".



Original source details: GELFAND-P000563

Undeciphered passage. The remaining words in the crossed-out line are illegible.

$$\frac{\delta}{\delta u}\left\{\sum c_n I_n\right\} = d \quad (= const).$$

. . . . . .

GELFAND-P000564

———— ——

23/IX – 74 Gelfand Sem.

On the work of Gabrielov, I. Gelfand and Losik on combinatorial topology.

It is interesting to do it analytically. ~~$\int$ [...]~~



Original source details: GELFAND-P000564

Uncertain reading. A mark with two loops begins the crossed-out group. It provisionally reads $\int$.



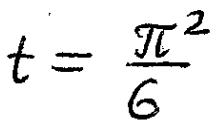
Undeciphered passage. Oblique strokes obscure the remaining marks. They are illegible.

Original source details: GELFAND-P000564

$$\int_{1-x}^{x} \left[\frac{\log|t|}{1-t} + \frac{\log|1-t|}{t} \right] dt$$

cannot be evaluated in elem. funcs.. But:

$$\int_0^1 \frac{\log|1-t|}{t}\, dt = \frac{\pi^2}{6} \quad \left(= \int_0^1 \sum_{n=1}^{\infty} \frac{t^{n-1}}{n} = \sum_{n=1}^{\infty} \frac{1}{n^2} \right).$$

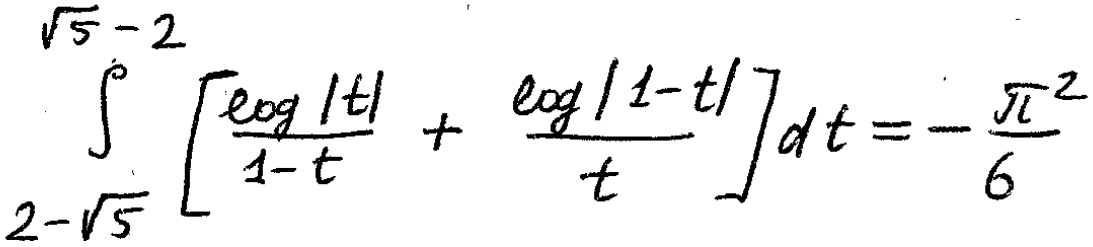


Uncertain reading. There is no minus sign before $\pi^2/6$. The middle term has no written differential.

Original source details: GELFAND-P000564

One can evaluate

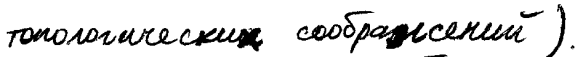
$$\int_{2-\sqrt{5}}^{\sqrt{5}-2} \left[\frac{\log|t|}{1-t} + \frac{\log|1-t|}{t} \right] dt = -\frac{\pi^2}{6}$$



Original source details: GELFAND-P000564

Uncertain reading. This line shows a minus sign before $\pi^2/6$. The limits read $2-\sqrt{5}$ and $\sqrt{5}-2$.

(from topological considerations).



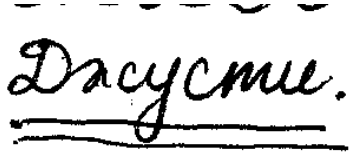
Uncertain reading. Both Russian words contain overwritten letters. The earlier strokes are illegible.

Original source details: GELFAND-P000564

They occur in a finite combinatorial problem.

Graeme Segal — all of topology.

Giusti. On singularities. (On regularity of ~~[...]~~ solv. of nonlin. syst.).



Uncertain reading. The name in the Russian handwriting provisionally reads “Джусти”. The middle strokes run together.

Original source details: GELFAND-P000564

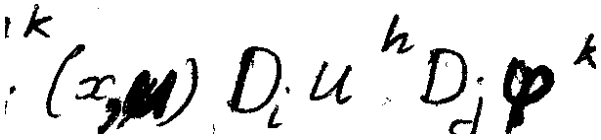


Undeciphered passage. A short entry before “solv.” in the English parenthesis is heavily crossed out and illegible.

Original source details: GELFAND-P000564

Symplest case:

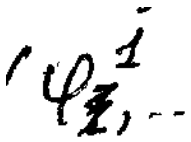
$$\sum_{i,j=1}^{n}\sum_{h,k=1}^{N}\int a_{ij}^{hk}(x,u)D_iu^hD_j\varphi^k\,dx=0$$



Uncertain reading. There are overwritten marks in the argument (x,u) and before the superscript k. The current readings u and φ are provisional.

Original source details: GELFAND-P000564

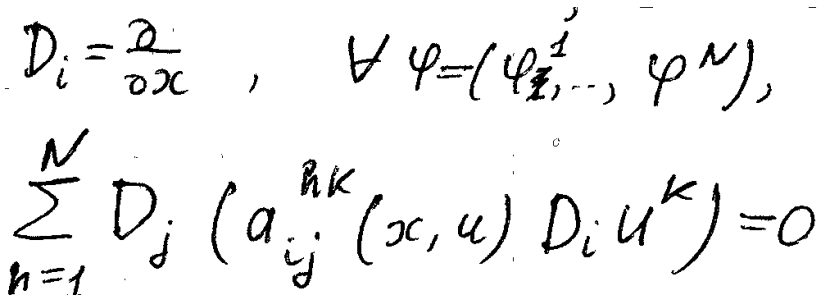
$$D_i=\frac{\partial}{\partial x},\qquad \forall\varphi=(\varphi^1,\dots,\varphi^N),\qquad \varphi_i\in W_0^{1,2}(\Omega)$$



Uncertain reading. There is a heavy subscript mark below φ^1. It may be 2, i or z. Its reading remains unresolved.

Original source details: GELFAND-P000564

$$\sum_{i,j=1}^{n}\sum_{h=1}^{N}D_j\big(a_{ij}^{hk}(x,u)D_iu^k\big)=0\qquad(k=1,2,\dots,N)$$



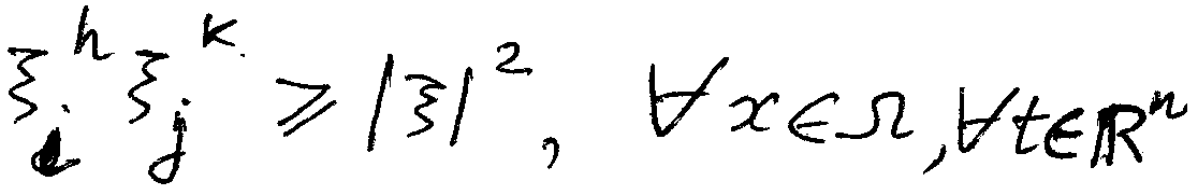
Original source details: GELFAND-P000564

Uncertain reading. The written denominator of D_i is ∂x. In the sum over h the visible superscript of u is k.

$$u = (u^1, \dots, u^N).$$

Ellipticity:

$$\sum_{ijhk} a_{ij}^{hk}(x,t)\xi_i^h \xi_j^k \geqslant |\xi|^2, \qquad \forall x \in \Omega,\ \forall t \in \mathbb{R}^n$$



Original source details: GELFAND-P000564

Uncertain reading. The first subscript on ξ provisionally reads i. The final dimension reads n.

GELFAND-P000565

$$\sum_{i,j=1}^{n} D_j\big(A_{ij}(x,u)D_i u\big) = 0, \qquad \begin{matrix} u \text{ — a vector} \\ A_{ij} \text{ — a matrix.} \end{matrix}$$

$N = 1 \Rightarrow \forall$ solution is regular

or $n \leqslant 2$ $\quad \forall u \in W^{1,1}$ is regular (depende on reg. of A-coefficients)



Uncertain reading. The sign between n and 2 has been written over. The current reading $\leqslant$ is provisional.

Original source details: GELFAND-P000565



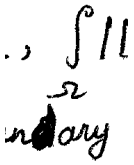
Uncertain reading. The mark before "coefficients" has been written over. It provisionally reads "A". The earlier strokes are illegible.

Original source details: GELFAND-P000565

$$A \in C^\infty \quad \Rightarrow \quad u \in C^\infty \quad \text{etc.}$$

$$W^{1,2}, \qquad u \in L_2, \qquad \int_\Omega |Du|^2\, dx < +\infty$$

$W_0^{1,2}$ — 0 boundary values.



Uncertain reading. The "d" in "boundary" has been written over. Above the word, Ω is below the integral in the previous line.

Original source details: GELFAND-P000565

$n = N \geqslant 3$ falls.



Uncertain reading. The digit “3” has been written over. The earlier strokes beneath it remain unresolved.

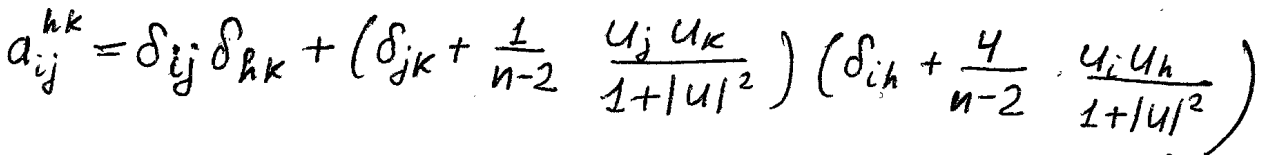
Original source details: GELFAND-P000565



Uncertain reading. The English word has two tall loops with no dot over the first. It is transcribed as “falls”.

Original source details: GELFAND-P000565

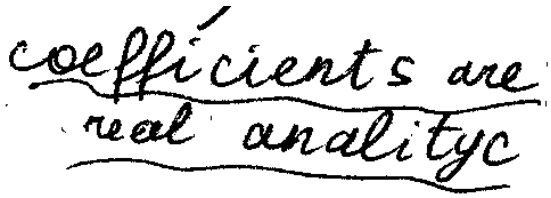
$$a_{ij}^{hk} = \delta_{ij}\delta_{hk} + \left(\delta_{jk} + \frac{1}{n-2}\frac{u_j u_k}{1+|u|^2}\right)\left(\delta_{ih} + \frac{4}{n-2}\frac{u_i u_h}{1+|u|^2}\right)$$



Original source details: GELFAND-P000565

Uncertain reading. The written factors preceding the fractions are $1/(n-2)$ and $4/(n-2)$, in that order.

$u_i = x_i/|x|$ — solution. coefficients are real analityc



Uncertain reading. The original English spelling “analityc” is retained.

Original source details: GELFAND-P000565

$n = 2$ $\forall$ sol. is regular

$n = 3,\ N = 2$ — there is an example of non regular solution.

For Dirichlet problem. $u_i = x_i$ when $|x| = 1$. $\exists!$ solution and it is not regular!

Suppose a_{ij}^{hk} are continuous

Then (1) $\forall$ solution u of system is regular in an open set $\Omega_0 \subset \Omega$



Uncertain reading. An entry in parentheses is crossed out before “(2)”. The earlier number is illegible.

Original source details: GELFAND-P000565

(2). $K = \Omega - \Omega_0$ (K is closed in Ω) ~~a[...]~~

$$\Rightarrow \underset{H_{n-2}}{\text{meas}} K = 0$$



Uncertain reading. An initial “a” is visible in the crossed-out English word. The remaining letters are unresolved.

Original source details: GELFAND-P000565

$$\underset{\sim}{\dim K < n-2.}$$



Uncertain reading. The lower stroke of the sign in the dim K line is thickened. The visible outline is transcribed as $<$.

Original source details: GELFAND-P000565

(for order $m > 2$ $\quad$ $\dim K \leqslant n-2$).

~~There~~ Are ~~also~~ [inserted above] there variational problems with nonregular solutions?



Uncertain reading. The crossed-out words before and after “Are” provisionally read “There” and “also”. The inserted “there” is enclosed above the line.

Original source details: GELFAND-P000565

Problem of Gelfand. Devise an analogue of Hironaka’s resol. of sing. for elliptic eqs..

GELFAND-P000566

I. M. Gelfand recap of Dubrovin’s previous talk.

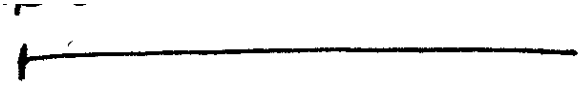


Diagram and typed description.

There is a small detached slanted stroke above the left end.

Original source details: GELFAND-P000566

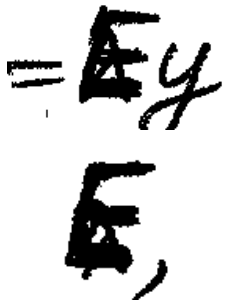
$$-y'' + u(x)y = Ey$$
$$u(x+T) = u(y)$$



Uncertain reading. The right-hand side clearly reads $u(y)$.

Original source details: GELFAND-P000566

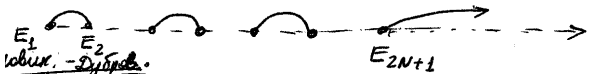
The spectrum — those E, for which there are bounded eigenfuncs.



Original source details: GELFAND-P000566

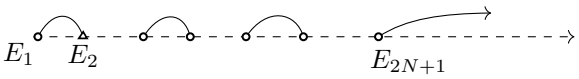
Uncertain reading. Both letters E are written over with heavy strokes. The earlier signs remain unresolved.

Novikov's problem: when are there finitely many forbidden zones?

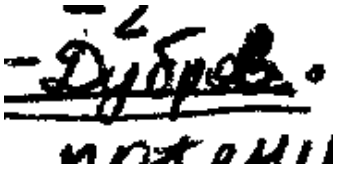


Original source details: GELFAND-P000566

Diagram and typed description.



Th. Novik.–Dubrov.



Original source details: GELFAND-P000566

Uncertain reading. A heavy stroke is written over the end of the abbreviated name "Dubrov." The earlier letter remains unresolved.

These are the potentials $\Leftrightarrow$ satisf. the nonlin. eq.

$$K(u, u', \dots, u^{(2N)}) = 0$$

(Higher Korteweg–de Vries eq.)

$$\begin{aligned} c(x, x_0, E): &\qquad c(x_0) = 1, \quad c'(x_0) = 0 \\ s(x, x_0, E): &\qquad s(x_0) = 0, \quad s'(x_0) = 1. \end{aligned}$$

These are analyt. funcs. of the parameter E.

$$\widehat{T}\Psi = \Psi(x+T), \qquad \widehat{T} = \begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{pmatrix} \qquad \det \|\widehat{T}\| = 1.$$

Bloch func.: $\Psi(x+T) = \lambda\Psi(x)$, $\qquad \Psi(x_0) = 1$

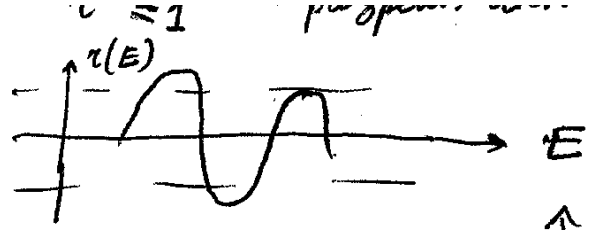
This is an eigenvector for $\widehat{T}$.

$$\lambda^2 - (\alpha_{11} + \alpha_{22})\lambda + 1 = 0,$$

$$\lambda^2 - 2\tau\lambda + 1 = 0, \qquad \tau(E) = \frac{\alpha_{11} + \alpha_{22}}{2}.$$

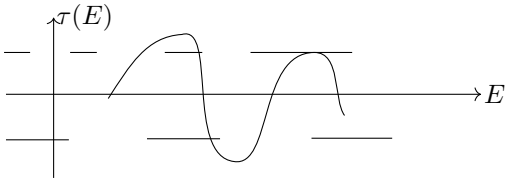
$E \in$ an allowed interv. $\Leftrightarrow \exists$ a Bloch func. with $|\lambda| = 1$.

$\tau^2 \leqslant 1$ — allowed int., $\tau^2 > 1$ — forbidden



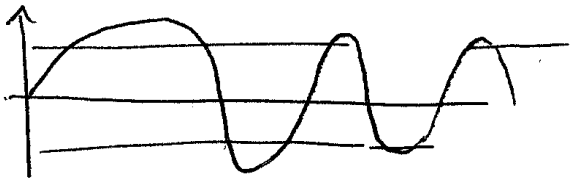
Original source details: GELFAND-P000566

Diagram and typed description.



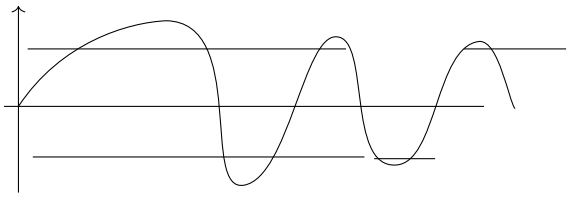
all $max \geqslant 1$ (this shape).

Correct fig.:



Original source details: GELFAND-P000566

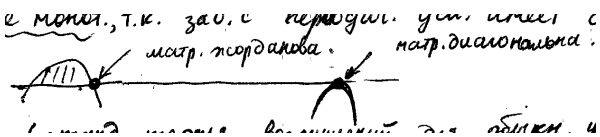
Diagram and typed description.



The forb. zones shrink.

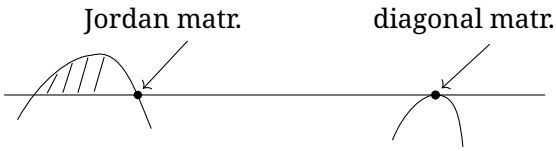
Important conclusion: if all the zones have merged, then, max and min are equal from some point on to 1.

They cannot, since the problem with periodic conds. has infinite spectrum (there $\tau = 1$).



Original source details: GELFAND-P000566

Diagram and typed description.



(stand. perturbation theory for ordinary eqs.).

$$\tau^2(E) = 1 \quad \text{has a mult. root} \quad \Rightarrow \quad \alpha_{12} = \alpha_{21} = 0.$$



Uncertain reading. The first digit of the subscript on α_{12} has been written over. The visible reading is 1.

Original source details: GELFAND-P000566

$\alpha_{12}(E)$ and $\alpha_{21}(E)$ have a zero at the same points where $\tau^2(E) - 1$ has multiple zeros.
Consid. the Bloch func. $\Psi(x, x_0, E)$: $\Psi(x, x_0, E)$, $\Psi(x_0) = 1$.
there are 2 Bloch funcs. $\forall E$ (since the eq. is quadr. in λ).
it depends analyt. on E (two-valued).

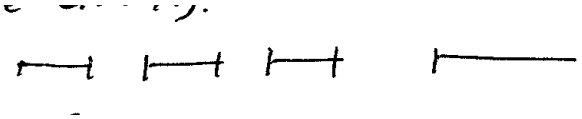


Diagram and typed description.

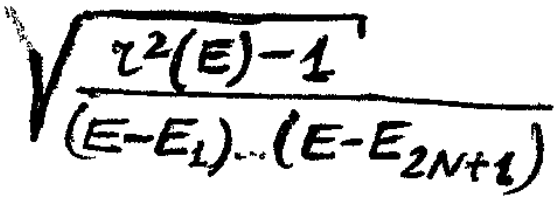


Original source details: GELFAND-P000566

$\tau^2(E) - 1$ has a zero at $E_1, \ldots, E_{2N+1}$. The rest are double.

GELFAND-P000567

$$\sqrt{\frac{\tau^2(E) - 1}{(E - E_1) \cdots (E - E_{2N+1})}} \quad \text{— single-valued func.}$$



Uncertain reading. The top stroke of the radical ends over the numerator. The bottom of the radical descends to the denominator. Its scope remains ambiguous.

Original source details: GELFAND-P000567

Consid. the eq.

$$-\Psi'' + u(x)\Psi = E\Psi, \qquad \Psi(x+T) = \lambda\Psi(x)$$
$$u(x+T) = u(x)$$

Let $\chi = -i\,\Psi'/\Psi$.

1°. χ does not depend on x_0, $\quad \chi = \chi(x, E)$.

2°. $\chi(x+T) = \chi(x)$.

3°. $-i\chi' + \chi^2 + u - E = 0$.

$\chi = \chi_R + i\chi_I$ in an allowed band — then analyt. continuation in E.

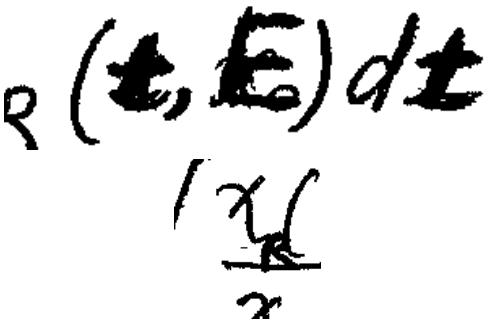
$$4^\circ. \quad \chi_I = \frac{1}{2}\,\chi_R'/\chi_R.$$

$$5^\circ. \quad \chi(x, E) \sim k + \sum_{n=1}^{\infty} \frac{\chi_n(x)}{(2k)^n}, \qquad E = k^2$$
$$\chi_n = P_n(u, u', \ldots, u^{(n)}) \text{ — a polynomial}$$

(det. successively from the asympt. equality)

$$4^\circ \Rightarrow 6^\circ. \quad \chi_R(x, E) \sim k + \sum_{n=0}^{\infty} \frac{\chi_{2n+1}(x)}{(2k)^{2n+1}}$$

$$7^\circ. \quad \Psi(x, x_0, E) = \sqrt{\frac{\chi_R(x_0, E)}{\chi_R(x, E)}} \cdot \exp\left[i \int_{x_0}^{x} \chi_R(t, E)\, dt\right].$$



Uncertain reading. The subscript of χ_R in the numerator has been written over. In the exponent, t, E and the final t of dt have been written over. The earlier strokes remain unresolved.

Original source details: GELFAND-P000567

$$7^\circ \Rightarrow 8^\circ. \quad |\Psi|^2 = \chi_R(x_0, E)/\chi_R(x, E) \qquad \underline{\text{in an allowed band}}$$

9°. ~~[...]~~



Uncertain reading. An entry before λ is heavily crossed out. Some strokes resemble an equals sign and $p(E)$. The exact reading remains unresolved.

Original source details: GELFAND-P000567

$$\lambda = e^{ip(E)} \quad \Rightarrow \quad p(E) = \int_{x_0}^{x_0+T} \chi_R(x,E)\,dx + 2\pi n.$$

The integral — a functional of u with an asympt. expansion. Its Euler eq. — KdV eqs., Allowed and forb. zones — where $|\lambda| = 1$



Uncertain reading. An earlier number with a superscript circle is crossed out to the left of "10°". A vertical stroke and a loop are visible. The exact digits remain unresolved.

Original source details: GELFAND-P000567

$$10^\circ. \quad \frac{\delta p(E)}{\delta u(x)} = -\frac{1}{2}\frac{1}{\chi_R(x,E)}$$

(left-hand side of the Euler eq. for KdV).

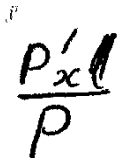
$$11^\circ. \quad \chi(x,E) = \frac{\sqrt{1-\tau^2(E)}}{\alpha_{21}(x,E)} + \frac{i}{2}\frac{\alpha_{11}(x,E)-\alpha_{22}(x,E)}{\alpha_{21}(x,E)}$$

Rough idea: eigenvalue obtained from the quadr. eq.

$$12^\circ. \quad \text{Let} \quad R(E) = \prod_1^{2N+1}(E-E_j). \qquad \text{Then}$$

$$\chi(x,E) = \frac{\sqrt{R(E)}}{P(E,x)} - \frac{i}{2}\frac{P'_x}{P},$$

where P — a polynomial in E of degree N.



Uncertain reading. A heavy stroke follows P'_x in the numerator. It resembles an opening parenthesis. Its meaning remains unresolved.

Original source details: GELFAND-P000567

GELFAND-P000568

$$P(E,x)=\prod_{j=1}^{N}(E-\gamma_j(x)).$$

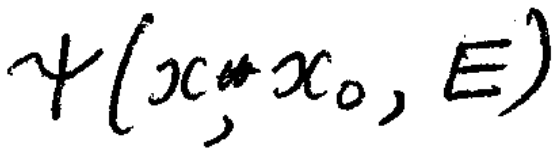
G.'s problem. Let R and P — entire funcs. of the 1st order of growth. Then corresponding to them are potentials $u(x)$. Which ones?

class of potentials (not necessarily periodic). ____ · ____

Dubrovin.

Γ_N — Riem. surf. of $\sqrt{R(E)}$, $\chi(x,E)$ — an algebr. func. on this surface

Theorem: $\Psi(x,x_0,E)$ a single-valued func. on Γ_N, merom. on $\Gamma_N\setminus\infty$, has N zeros and N poles.



Uncertain reading. A comma follows the first x. A small mark is heavily crossed out above it. The earlier reading remains unresolved.

Original source details: GELFAND-P000568

Poles at $E=\gamma_i(x_0)$, zeros above $E=\gamma_i(x)$.

GELFAND-P000569

30/IX – 74. Gelfand Sem.

Polyakov. On elementary particles. Yang–Mills fields. (difference analogue).

Graeme Segal. A work of Dusa Mc Duff on config. space.

$$\mathbb{R}^n,\qquad C_k(\mathbb{R}^n)=\{\text{subsets of }\mathbb{R}^n\text{ with }k\text{ points (no ordered)}\}.$$

From a top. point of view it is not quite trivial.

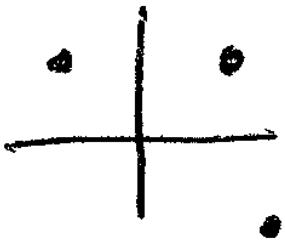
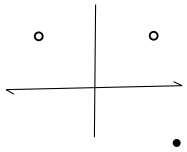


Diagram and typed description.



Original source details: GELFAND-P000569

~~[...] f_S~~



Undeciphered passage. The left part of the crossed-out entry is illegible.

Original source details: GELFAND-P000569



Original source details: GELFAND-P000569

Uncertain reading. The end of the crossed-out entry provisionally reads f_S.

constr. a map $f_S : \mathbb{R}^n \longrightarrow \mathbb{R}^n \cup \infty$

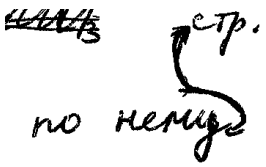
$S \subset \mathbb{R}^n$ from it

finite.



Original source details: GELFAND-P000569

Uncertain reading. The abbreviation before "map" reads "стр." An arrow points up to it.



Original source details: GELFAND-P000569

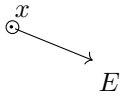
Uncertain reading. The end of the word merges with the arrow. It provisionally reads "from it", possibly "after the German".

E — field strength (electr.), produced by unit charges at points $x \in S$. We get a vector $E \in \mathbb{R}^n \cup \infty$



Original source details: GELFAND-P000569

Diagram and typed description.



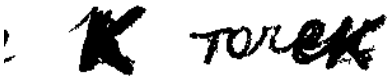
At the points $x \in S$ it will be ∞. $\infty \longrightarrow 0$

$$f_S : \mathbb{R}^n \cup \infty \longrightarrow \mathbb{R}^n \cup \infty$$

We get a map

$$C_k(\mathbb{R}^n) \longrightarrow \mathrm{Map}_k(S^n, S^n). \qquad (\text{map of degree } k).$$

(When far away, all k points are perceived as one).



Original source details: GELFAND-P000569

Uncertain reading. The k and the end of the Russian word for "points" are written over. The earlier strokes are illegible.

A point corresp. to a long vector — near one of the k charges
— and there are such points.



Uncertain reading. A letter in the middle of the Russian word for “such” is heavily overwritten. The earlier strokes are illegible.

Original source details: GELFAND-P000569

$C_k(\mathbb{R}^n)$ — a good approx. of $\mathrm{Map}_k(S^n, S^n)$!! (in a certain sense)
A good approx. for large k.

$$H_i(C_k(\mathbb{R}^n)) \underset{\cong}{\longrightarrow} H_i(\mathrm{Map}_k)$$

for $i \ll k$ (there are many proofs, including Segal’s)



Uncertain reading. Below the arrow is a wave over two horizontal strokes, in the form $\cong$.

Original source details: GELFAND-P000569

Inventiones

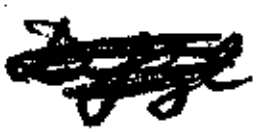


Uncertain reading. The underlined word reads “Inventiones”. The final letters are joined.

Original source details: GELFAND-P000569

GELFAND-P000570

A generalization of this thm. will be described (due to ~~[...]~~ Dusa Mc Duff)



Undeciphered passage. A short entry before the name “Dusa Mc Duff” is crossed out and illegible.

Original source details: GELFAND-P000570

Fuchs. $C_k(\mathbb{R}^n)$ gives various classifying spaces.



Uncertain reading. The initial letter of “Fuchs” is written over with a heavy stroke. The earlier strokes remain unresolved.

Original source details: GELFAND-P000570

2 kinds of questions arise:

The right side is indep. of k ($\approx$), on the left pass to the limit in k.



Original source details: GELFAND-P000570

Uncertain reading. The middle of the Russian word for “limit” is written over. The earlier strokes are illegible.

(homot. eq.)

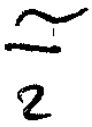
$C_k^{\pm}(\mathbb{R}^n)$ finite subsets with charges of diff. signs

topology — if 2 oppositely charged particles approach, they disappear.

Many components (total charge), $-\infty < k < +\infty$

Does not depend strongly on k.

$$C_k^{\pm}(\mathbb{R}^n) \underset{2}{\simeq} \mathrm{Map}_k(S^n, S^n)$$



Original source details: GELFAND-P000570

Uncertain reading. A separate mark in the shape of 2 is below $\simeq$. Its meaning remains unresolved.

a very good approx. (hypothesis). or



Original source details: GELFAND-P000570

Uncertain reading. Heavy crossing strokes follow “hypothesis”. The provisional reading is “or”.

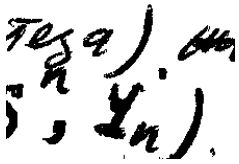
There is a space Y_n, $\dim Y_n = 2n$,

$$C_k^{\pm}(\mathbb{R}^n) \simeq \mathrm{Map}_k(S^n, Y_n).$$



Original source details: GELFAND-P000570

Uncertain reading. The upper stroke of the relation is written over. The visible wave over a line is transcribed as $\simeq$.



Original source details: GELFAND-P000570

Uncertain reading. The Y in the final Y_n is written over. The earlier strokes are illegible.

$C_k(M)$ M — a manifold

What in place of Map(S^n, S^n)?

$$C_k(M) \longrightarrow \Gamma(T_M^{\text{comp}}),$$

∞ is added in each fibre. (all fibres compactified).

At [...] infinity one can take 0 (with comp. supp.)



Uncertain reading. A small loop-shaped mark is crossed out before “infinity”. Its reading remains unresolved.

Original source details: GELFAND-P000570

$$C_k(M) \longrightarrow \Gamma_k^{\text{cpt}}(T_M^{\text{comp}})$$

induces a homol. isom. for $i \ll k$.

(The electr. field is taken to have finite range — then everything is constr.)

The proof method is of greater interest than the assertion.

$U \subset M$ a small piece of M. $C_k(U)$ [...] $C_k(M)$.



Uncertain reading. A sign between $C_k(U)$ and $C_k(M)$ is heavily crossed out. Its reading remains unresolved.

Original source details: GELFAND-P000570

We would like to obtain $C_k(M)$, by joining $C_k(U)$.

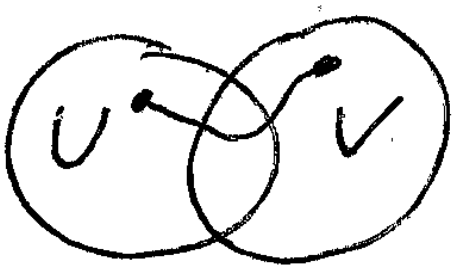
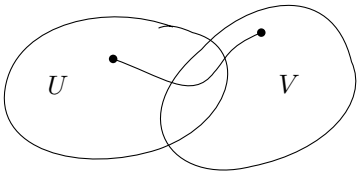


Diagram and typed description.



Original source details: GELFAND-P000570

$$C(M) \underset{\text{discontinuous}}{\longrightarrow} C(U) \times C(V)$$



Uncertain reading. “Discontinuous” is written below the formula and above the next line.

Original source details: GELFAND-P000570

Let U, V — manifolds with boundary.

$\overline{C}(U)$ — the quotient space of $C(U)$

$$S_1 \sim S_2 \quad \Leftrightarrow \quad S_1 \cap \overset{\circ}{U} = S_2 \cap \overset{\circ}{U}$$

(points on the boundary disappear)

Thm. U — a manif. with boundary (nonempty)

$$\Rightarrow \quad \overline{C}(U) \underset{\simeq}{\longrightarrow} \Gamma(TU^{\text{comp}}) \qquad \text{(homotopic equiv.)}.$$

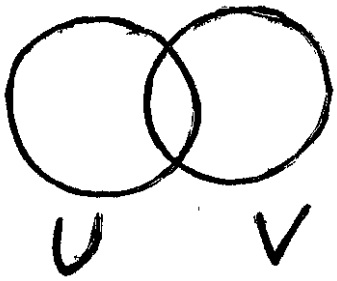
Proof. Case 1. $U =$ the unit disk.

Left side: $C_\varepsilon(U)$ — configurs. with distance $> \varepsilon$. Configurs. of $\leqslant 1$ particles remain.

right side $\quad \text{Map}(U, S^n) \simeq S^n$

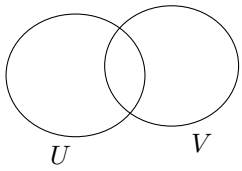
Radial stretch. $\longrightarrow S^n$ (pushed out to the boundary)

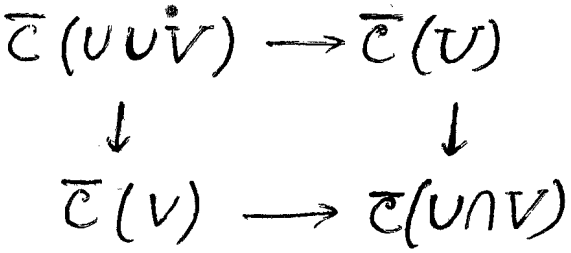
GELFAND-P000571



Original source details: GELFAND-P000571

Diagram and typed description.





Original source details: GELFAND-P000571

Diagram and typed description.

$$\begin{array}{ccc} \overline{C}(U \cup \dot{V}) & \longrightarrow & \overline{C}(U) \\ \downarrow & & \downarrow \\ \overline{C}(V) & \longrightarrow & \overline{C}(U \cap V) \end{array}$$

A dot stands above V in the upper left. Its meaning remains unresolved. The lower arrow points right.

there are well-defined maps and if 2 points from $\overline{C}(U)$ and $\overline{C}(V)$ det. one in $\overline{C}(U \cap V)$, then ~~this~~ det. a point in $\overline{C}(U \cup V)$.



Uncertain reading. "This" is crossed out after "then".

Original source details: GELFAND-P000571

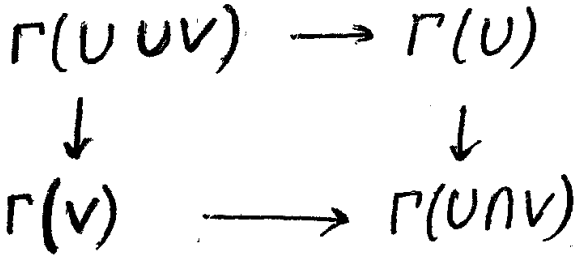
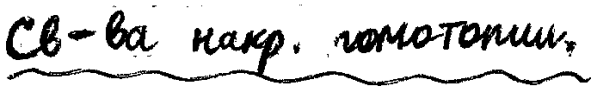


Diagram and typed description.

$$\begin{array}{ccc} \Gamma(U \cup V) & \longrightarrow & \Gamma(U) \\ \downarrow & & \downarrow \\ \Gamma(V) & \longrightarrow & \Gamma(U \cap V) \end{array}$$

Original source details: GELFAND-P000571

Props. of covering homotopy.



Uncertain reading. The middle Russian abbreviation reads "накр.".

Original source details: GELFAND-P000571

the analogous thm. on $C_k(M)$ is omitted.

Next time: Polyakov or Dubrovin.

GELFAND-P000572

7/X–74. Gelfand Sem.

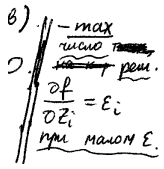
Kushnirenko. $f(z_1, \dots, z_n) = \sum a_m z^m$

0 — sing. point. (series without const. and linear terms).

Milnor number $\mu(f)$ = multiplicity of the zero sol.

$$\frac{\partial f}{\partial z_1} = 0, \dots, \frac{\partial f}{\partial z_n} = 0.$$

— max number [...] [...] sols.
$\frac{\partial f}{\partial z_i} = \varepsilon_i$
for small ε.

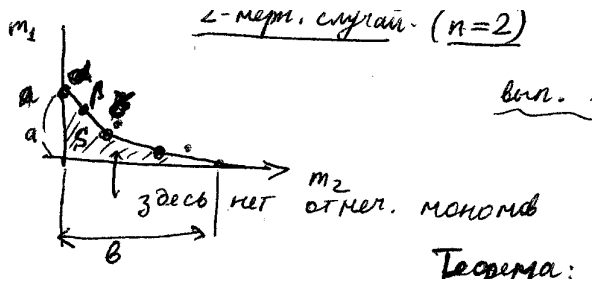


Original source details: GELFAND-P000572

Undeciphered passage. Two groups in the right margin are crossed out after “number” and before “sols.”. Remnants of strokes are visible. The full reading remains unresolved.

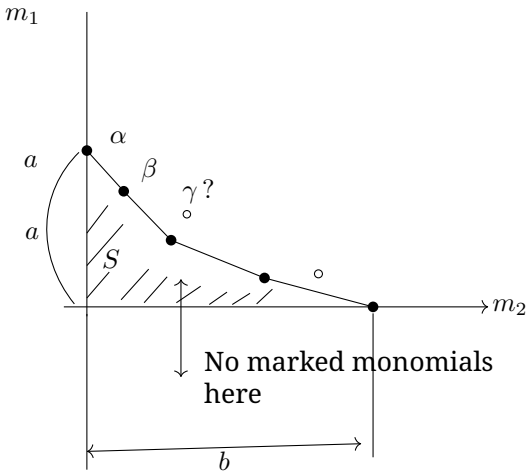
Problem: compute $\mu(f)$ from the coeffs. of the Taylor series.

2-dim. case. ($n = 2$)



Original source details: GELFAND-P000572

Diagram and typed description.



The third label is written over. Its shape and the formula below suggest γ. The reading remains provisional. The two small marks above the polygonal line are retained.

conv. polygonal line

$$\nu(f) = 2S - a - b + 1.$$

Theorem: 1) $\mu(f) \geqslant \nu(f)$

2) $\mu(f) = \nu(f)$, if certain nondegeneracy conds. hold

$\beta^2 \neq 4\alpha\gamma$ (α, β, γ — coefficients of the resp. monomials).

(in the gen. case $\neq 0$ discriminant of the left edge.)

↘ on the edge essentially a polynomial of 1 var.

$n = 3$.

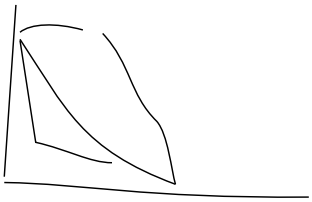
$$\nu(f) = 3!V - 2!(S_{12} + S_{13} + S_{23}) + 1!(L_1 + L_2 + L_3) - 1.$$

S_{12} — area obtained on the plane.



Original source details: GELFAND-P000572

Diagram and typed description.



The contours and their spatial relation are schematic in the manuscript.



Original source details: GELFAND-P000572

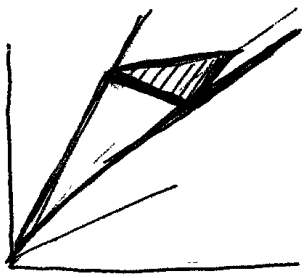
Uncertain reading. A short detached oblique stroke is visible between the formula for ν and the drawing. Its function is unresolved.

Theorem: 1) $\mu(f) \geqslant \nu(f)$

2) nondegen. cond. $\Rightarrow$ equal-ty.

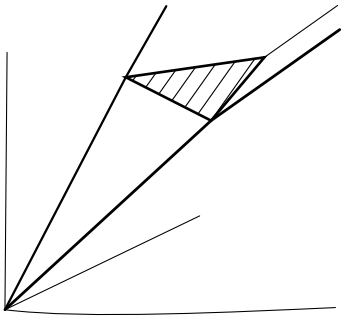
(holds almost always)

GELFAND-P000573



Original source details: GELFAND-P000573

Diagram and typed description.



The hatched region lies between the upper rays. Heavy and retraced strokes are drawn schematically. The precise directions of the extensions are unresolved.

$$z_i \frac{\partial f_\Delta}{\partial z_i}$$

f_Δ — corresp. to the monomials of the face.

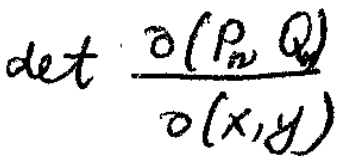
$$\dim_{\mathbb{C}} \mathbb{C}[P(\Delta)] / \left(z_i \frac{\partial f}{\partial z_i} \right) < \infty.$$

nondegen. cond.

Proof: commut. algebra. Use that $\mathbb{C}[P(\Delta)]$ — a Cohen–Macaulay ring (Hochster).

Arnold.

$$\sum_{x,y:\begin{cases} P_n(x,y)=0 \\ Q_n(x,y)=0 \end{cases}}^{n^2} \frac{R_{n-4}(x,y)}{\det \frac{\partial (P_n\, Q)}{\partial (x,y)}} = 0.$$



Original source details: GELFAND-P000573

Uncertain reading. The numerator of the Jacobian shows Q after P_n. No distinct comma is visible, and any index on Q is obscured by the tail and parenthesis. The summation conditions separately show Q_n.

$P_n(x,y)$, $Q_n(x,y)$ — 2 polynomials.

(R — an arb. polynomial of deg. $\leq n-4$)

$\sum$ over all zeros.

Polyakov. Consider $\mathbb{Z}^2$ and a fn. of an oriented edge connecting adjacent vertices.



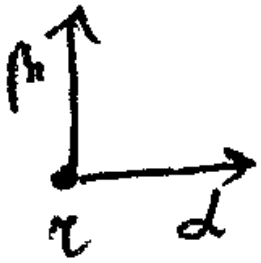
Source raster detail: GELFAND-P000573

Icon visible in the source raster. A yellow icon with three strokes is visible above the word for “oriented” in the raster. Its text is absent from the image and its origin is unresolved.

Yang–Mills fields.

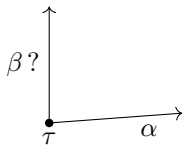
$$B_{\tau,\alpha}, \quad \tau \in \mathbb{Z}^2, \quad \alpha \text{ — dir.}$$

a fn. with vals. in SU(2) (or another compact gr.).



Original source details: GELFAND-P000573

Diagram and typed description.



The label by the vertical arrow resembles β, but its shape is ambiguous. The horizontal arrow is labelled α.

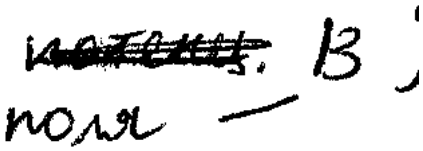
$$F_{\vec{\tau},\alpha\beta} = \quad \text{circul.} = B_{\tau\alpha} B_{\tau+a_\alpha,\beta} B^{-1}_{\tau+a_\beta,\alpha} B^{-1}_{\tau\beta}.$$



Original source details: GELFAND-P000573

Manuscript excerpt. The top, left, bottom and right arrows point left, down, right and up, respectively.

(field strength of ~~[...]~~ B). $\hookleftarrow$ field

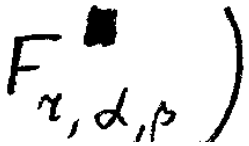


Original source details: GELFAND-P000573

Uncertain reading. The earlier word before B is crossed out, provisionally "of the contour". The inserted word "field" appears below with a connecting stroke.

Energy

$$E = \sum_{\tau,\alpha,\beta} \chi(F_{\tau,\alpha,\beta}) \qquad Z = \int \prod dB_{\tau,\alpha}\, e^{-\frac{1}{T}E[B]}.$$



Original source details: GELFAND-P000573

Uncertain reading. A dense dark patch is visible above F. The underlying sign and its function are unresolved.

χ — trace.

Correl. fns.:

$$\Pi(C) = Z^{-1} \int \prod dB_{\tau\alpha}\, \chi \left[\prod_{(C)} B_{\tau\lambda} \right] e^{-\frac{1}{T}E(B)}.$$

$\alpha, \beta = 1$ or 2

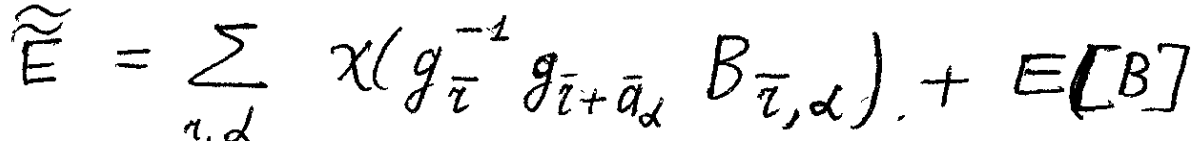
C — contour.

Lattice of size $N \times N$.

The planar prob. is solved exactly.

$$\widetilde{E} = \sum_{\tau,\alpha} \chi(g^{-1}_{\vec{\tau}} g_{\vec{\tau}+\vec{a}_\alpha})$$

$$\widetilde{\widetilde{E}} = \sum_{\tau,\alpha} \chi(g^{-1}_{\bar{\tau}} g_{\bar{\tau}+\bar{a}_\alpha} B_{\bar{\tau},\alpha}). + E[B].$$



Original source details: GELFAND-P000573

Uncertain reading. In the double-tilde line B follows the second g within the parentheses with no visible comma between them. The τ index on B has a bar. There is a small separate stroke near the baseline before the plus sign.

Field th. = stat. mech. in $\mathbb{R}^n$ (Euclidean).

(Fourier transf. in time)

Sinai.

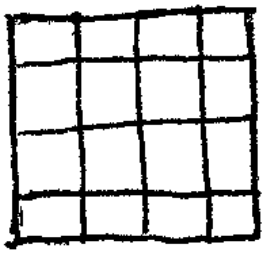
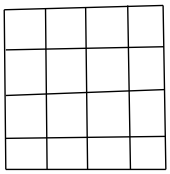


Diagram and typed description.



Original source details: GELFAND-P000573

V — finite volume.

$$E(\{B_{\tau\alpha}\}) = \sum \operatorname{tr}\left(B_{\tau\alpha} B_{\tau+a_\alpha,\beta} B^{-1}_{\tau+a_\beta,\alpha} B^{-1}_{\tau,\beta}\right)$$

$$\underline{\underline{\mathcal{M} e^{-\beta E} = Z(\beta, V)}}$$

Calculated for the 2-dim. case.

$$\ln Z(\beta, V) = V \ln I_0(\beta) \qquad (I_0(\beta) \text{ — Bessel fn.})$$

GELFAND-P000574

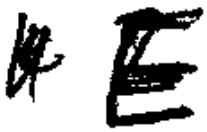
$$B_{\tau\alpha} = e^{iA_{\tau\alpha}}$$



Uncertain reading. The first subscript on B and on A reads τ.

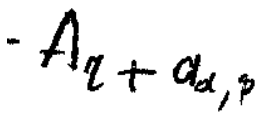
Original source details: GELFAND-P000574

$$\text{[...]}\ E = \sum \cos(A_{\tau\alpha} + A_{\tau+a_\alpha,\beta} - \ldots)$$



Uncertain reading. A short sign is densely crossed out before E. φ is possible, but the reading is unresolved.

Original source details: GELFAND-P000574



Uncertain reading. The second A under the cosine has $\tau + a_\alpha$, a comma and β in its subscript.

Original source details: GELFAND-P000574

$$\lim_{V\to\infty} \frac{\ln Z(\beta, V)}{V} = f(\beta)$$

does not depend on the choice of boundary conds.

$f(\beta)$ may have singularities — they are called phase transitions.

(≈ a rearrangement of typical configs. occurs).

$$p(\{B_{\tau,\alpha}\}) = Z^{-1} e^{-\beta E(\{B_{\tau\alpha}\})} \qquad \text{— Gibbs distrib.}$$

Correl. fn. — mean value of the circulation around the contour.
Phase trans. — in one phase ∼ area
in the other ∼ length.

GELFAND-P000575

~~[...]~~ 14/X–74. Gelfand Sem.



Uncertain reading. A short group is crossed out before the date. The Russian letters “Пo” are possible, but the strokes overlap.

Original source details: GELFAND-P000575

Graeme Segal.
Classifying space of category.
$\mathcal{C}$ — category, $|\mathcal{C}|$ is top. space
Objects — points
Morphisms — 1-dim symplex.
Compositions — ?

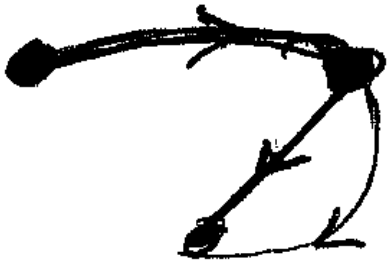
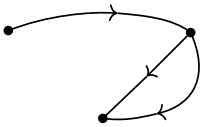


Diagram and typed description.



Original source details: GELFAND-P000575

we have a graph.
It describe not cat. [?]



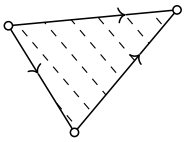
Uncertain reading. A low loop-like sign follows “cat.”. The reading $\mathcal{C}$ is tentative. Its shape differs from the clearly written letter above.

Original source details: GELFAND-P000575

(composition!)



Diagram and typed description.



Original source details: GELFAND-P000575

two-symplex — composition of two morphisms.

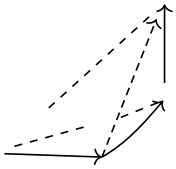
3 morph., and all compos. — 3-dim symplex.

which can be composed ↑



Original source details: GELFAND-P000575

Diagram and typed description.



The solid arrows and discernible thin strokes are drawn schematically. Some overlapping edges have unresolved directions.

etc.

We define $|\mathcal{C}|$.

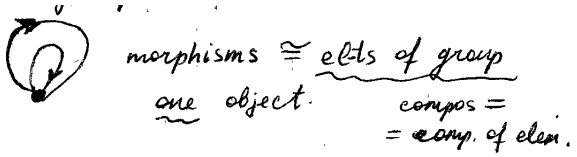
Diametr. opposed examples.

Partially ↓

Ordered sets.

≡ categ. with at most one morph. between two objects.

groups



Original source details: GELFAND-P000575

Diagram and typed description.



morphisms ≅ elts of group
one object.
compos =
= comp. of elem.

X top space, $\{U_\alpha\}_{\alpha\in S}$ — covering $U_\alpha \subset X$

open sets

We suppose:

Property:

$$U_\alpha \cap U_\beta = \bigcup_{\text{certain } \gamma} U_\gamma.$$

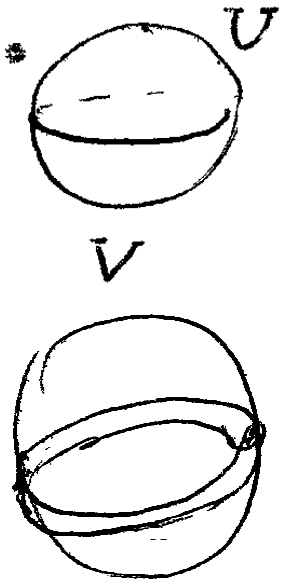
$$\alpha \le \beta \underset{\text{def}}{\iff} U_\alpha \subset U_\beta.$$

If all sets are contractible then $|\mathcal{C}| \underset{\text{hom. equiv.}}{\simeq} X$.

Add. intersections.

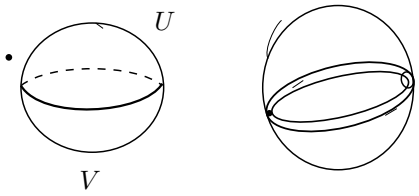
GELFAND-P000576

Example. $X = S^2$. $w = U \cap V$.

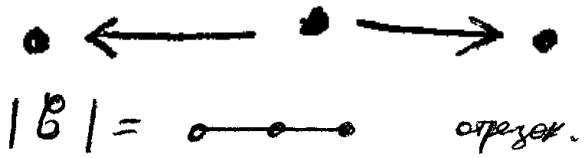


Original source details: GELFAND-P000576

Diagram and typed description.



A small mark is visible to the left of the upper outline. The band lines and side marks overlap in parts of the lower drawing.



Original source details: GELFAND-P000576

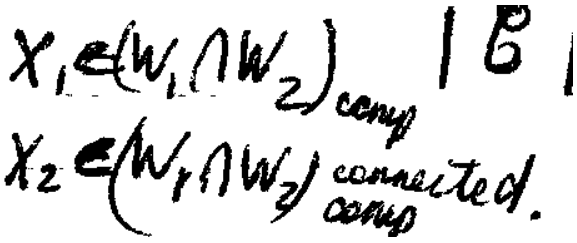
Diagram and typed description.



But W is not contractible.

$|\mathcal{C}| =$ segment.

$$w = W_1 \cup W_2.$$
$$x_1 \in (W_1 \cap W_2)_{\text{comp}}.$$
$$x_2 \in (W_1 \cap W_2)_{\substack{\text{connected} \\ \text{comp}}}.$$



Uncertain reading. “comp” is written to the right of the first parenthesis. “connected” with “comp” below follows the second.

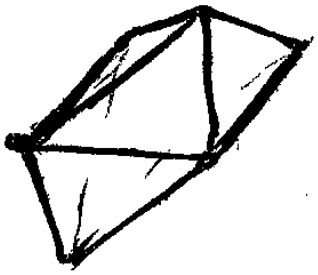
Original source details: GELFAND-P000576

~~oc?~~ — surface of octahedron.



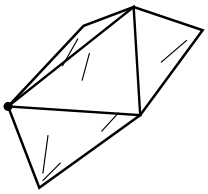
Uncertain reading. Two loop-like letters are visible under the cancellation, tentatively “oc”.

Original source details: GELFAND-P000576



Original source details: GELFAND-P000576

Diagram and typed description.



The redraw follows the clear outline and inner strokes. Some thin and superimposed edges are only partly discernible.

Ex. 2

V — finite dimensional vect. space ($\forall$ field).

$\mathcal{C}$ — cat of proper subspaces and inclusions.

$|\mathcal{C}|$ = "building" of V building

"immeuble"

Thm. (Solomon — Tits).

~~[...]~~ $|\mathcal{C}| \simeq$ wedge (bouquet) of spheres of dim $(n-2)$

$(n = \dim V)$,



Original source details: GELFAND-P000576

Uncertain reading. A cancelled group with a vertical and loops precedes $|\mathcal{C}|$. An earlier $|\mathcal{C}|$ is possible.

(quantity of spheres depends on field — infinite if the field is infinite).

Proof.

1. If ord. set has max elt then $|\mathcal{C}|$ is contractible.



Original source details: GELFAND-P000576

Diagram and typed description.



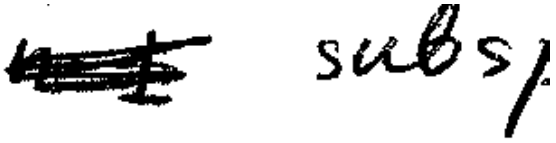
evidente.

max elements are ~~[...]~~ subsp. of codimension 1.



Original source details: GELFAND-P000576

Uncertain reading. A short abbreviation, possibly "vect", is crossed out before "subsp.".



Original source details: GELFAND-P000576

Undeciphered passage. Multiple horizontal strokes conceal the middle of the cancelled abbreviation.

$$|\mathcal{C}| = \bigcup_{H\text{—hyperplanes}} |\mathcal{C}_H|.$$

$\mathcal{C}_H$ — is the cat of subspace W, $0 \underset{\text{strict}}{\subset} W \subseteq H$.



Uncertain reading. The manuscript uses inclusion signs: $\subset$ annotated "strict" and $\subseteq$ with a lower bar.

Original source details: GELFAND-P000576

$|\mathcal{C}_H|$ is contractible.

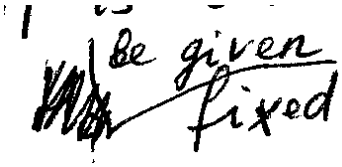
be given ↙

Let ~~[...]~~ fixed ~~hyperplane~~ line L.



Uncertain reading. A group between "Let" and "fixed" is crossed out, possibly "We".

Original source details: GELFAND-P000576

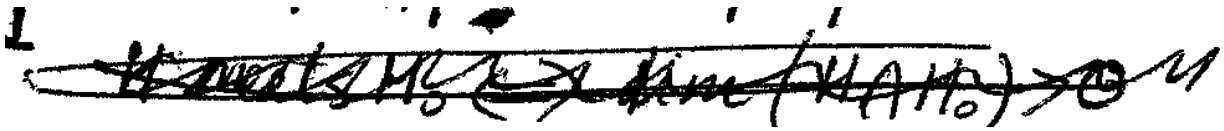


Undeciphered passage. Dense slanted strokes obscure the full group. "be given" is written separately above the line.

Original source details: GELFAND-P000576

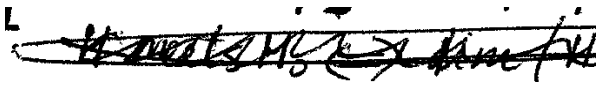
$$\text{Let} \quad Y = \bigcup_{H \supset L} |\mathcal{C}_H| \subset |\mathcal{C}|.$$

~~[...] $\Rightarrow$ dim$(W \cap H_0) > 0$ [...]~~



Original source details: GELFAND-P000576

Uncertain reading. The end of the cancelled line reads dim$(W \cap H_0) > 0$. The sign before "dim" resembles an implication arrow, and the final stroke may be Cyrillic "и".



Undeciphered passage. The opening and intervening parts of the cancelled line remain unresolved.

Original source details: GELFAND-P000576

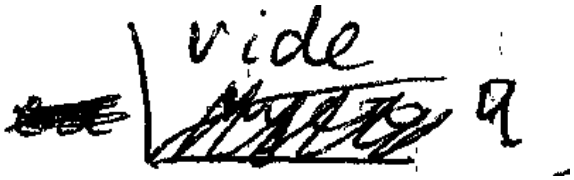
Y is contractible because

$$|\mathcal{C}_{H_1}| \underset{\text{contr.,}}{\cap} |\mathcal{C}_{H_2}| = \underset{\text{also contractible}}{|\mathcal{C}_{H_1 \cap H_2}|}.$$

GELFAND-P000577

If Y is covered by pieces Y_i such that all $Y_{i_1} \cap \ldots \cap Y_{i_k}$

is contractible (and not ~~[...]~~ vide ~~empty?~~ [?]) $\Rightarrow Y$ contractible.



Original source details: GELFAND-P000577

Uncertain reading. “vide” is above the crossed-out word, tentatively Russian “пусто”. A loop-like sign with a descender is to the right. Its reading is unresolved.



Original source details: GELFAND-P000577

Undeciphered passage. The first short word after “not” is crossed out. “et” is possible, but the full lettering is indistinct.

~~[...]~~



Original source details: GELFAND-P000577

Uncertain reading. The short cancellation below “is” begins with strokes that resemble “te”.



Original source details: GELFAND-P000577

Undeciphered passage. Dense crossing strokes obscure the ending.

in partial.

$$|\mathcal{C}| = Y \cup \bigcup_{H \not\supset L} |\mathcal{C}_H|.$$



Original source details: GELFAND-P000577

Uncertain reading. The union is indexed by $H \not\supset L$. The negated curved sign has no separate lower bar.

Lemma 1.

If $X = Y \cup Y_1 \cup \ldots \cup Y_k$

and $Y_?, Y_1, \ldots, Y_k$ contractible.

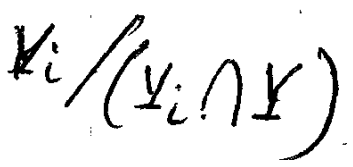


Original source details: GELFAND-P000577

Uncertain reading. A subscript on the first Y is crossed out. Its reading is unresolved.

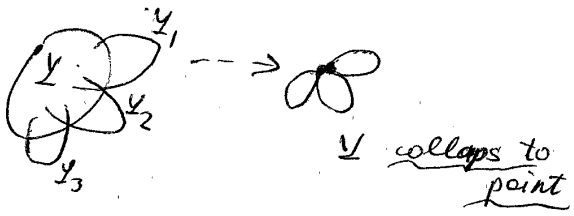
and $Y_i \cap Y_j \subset Y$, then

$$X \simeq \bigvee_{\text{wedge}} Y_i/(Y_i \cap Y)$$



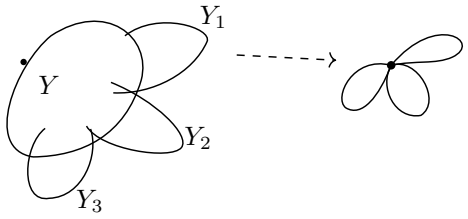
Original source details: GELFAND-P000577

Uncertain reading. The numerator resembles Y_i. The short stroke at its foot matches Y in the denominator. V_i remains a possible reading.



Original source details: GELFAND-P000577

Diagram and typed description.



Y collaps to point

Lemma 2.

Also if Z is contractible and $Z_0 \subset Z$ then

$$Z/Z_0 \simeq \underset{\text{suspension}}{SZ_0}$$

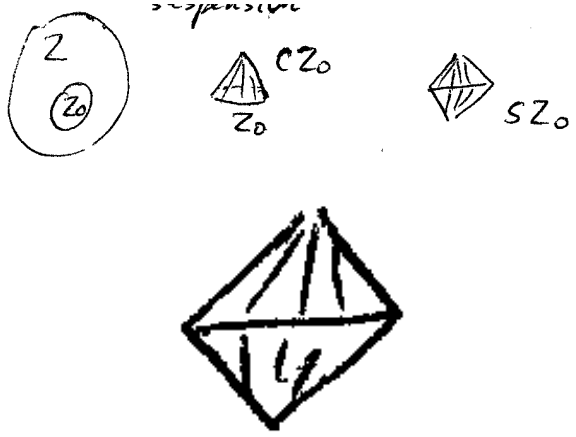
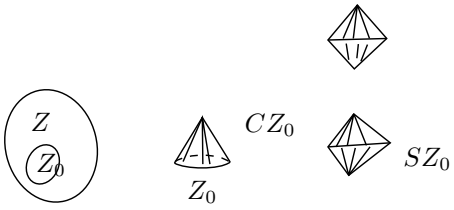


Diagram and typed description.



Original source details: GELFAND-P000577

$$\Rightarrow X \simeq \bigvee S(Y_i \cap Y)$$

Lemma 3.

$|\mathcal{C}_H| \cap Y =$ "building" of H.

and reduction by dimension.

dim $= 2$. (first step of induction).



Original source details: GELFAND-P000577

Diagram and typed description.

Lines

wedge of !! points ~~[...]~~



Original source details: GELFAND-P000577

Uncertain reading. "of" is visible in an outline after "wedge", with two vertical marks nearby. Their function is uncertain.

~~[...]~~



Original source details: GELFAND-P000577

Uncertain reading. Some strokes in the lower cancelled group resemble "wedge".



Original source details: GELFAND-P000577

Undeciphered passage. The lower inscription and the cancelled group after "points" are incompletely legible.

Suspension of wedge of spheres is also a wedge of spheres.

$H_{n-2}(|\mathcal{C}|;\mathbb{Z}) =$ Steinberg module of V.



Original source details: GELFAND-P000577

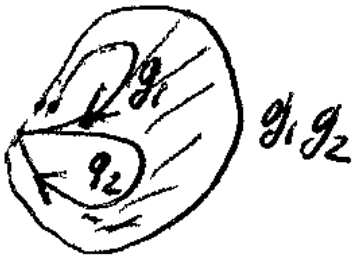
Uncertain reading. Typeset as $=$. The lower stroke is retraced, so $\equiv$ is a possible reading.

(repres. in subgrps. of a fin. group)

Other extreme case.

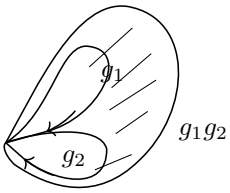
$\mathcal{C}$ of group G. $\quad |\mathcal{C}| = BG$ — classifying space of group

$\pi_1(|\mathcal{C}|) = G$. $\quad$ (evidently)



Original source details: GELFAND-P000577

Diagram and typed description.



The outline and loops follow the discernible strokes. Short, superimposed marks near the outline do not establish further directions.

Universal covering is contractible.

Let $\widetilde{\mathcal{C}}$ $\quad$ objects $=$ elts of G.

$\exists\, 1$ morph. $g \to g'$ for each pair of elts.

GELFAND-P000578

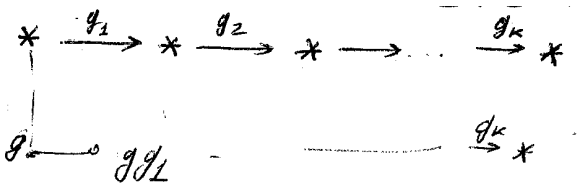
$|\widetilde{\mathcal{C}}|$ is contractible.

($\forall$ el-t is connected by line with unity 1).

$\widetilde{\mathcal{C}} \xrightarrow{\text{functor}} \mathcal{C}$ universal covering space.

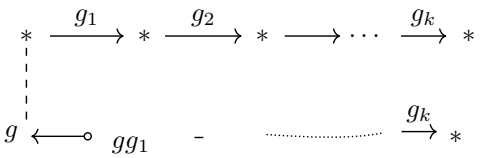
$g \mapsto *$

$g \to g' \mapsto g^{-1}g'$



Original source details: GELFAND-P000578

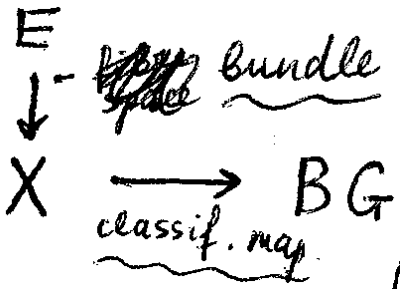
Diagram and typed description.



The short lower-left arrow points from a small loop towards g. An asterisk is visible at the right.

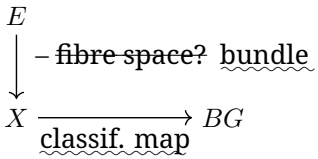
$|\widetilde{\mathcal{C}}| \to |\mathcal{C}|$ universal covering.

— · —



Original source details: GELFAND-P000578

Diagram and typed description.





Original source details: GELFAND-P000578

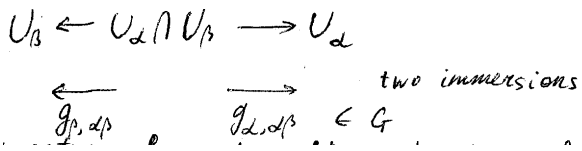
Uncertain reading. The crossed-out expression next to “bundle” tentatively reads “fibre space”.

$\{U_\alpha\}$ — covering, on U_α bundle is trivial.



Original source details: GELFAND-P000578

Uncertain reading. A heavy superimposed stroke follows “covering” before the comma. A separate character is uncertain.



Original source details: GELFAND-P000578

Diagram and typed description.

$U_\beta \longleftarrow U_\alpha \cap U_\beta \longrightarrow U_\alpha$

$\overleftarrow{g_{\beta,\alpha\beta}}$ $\overrightarrow{g_{\alpha,\alpha\beta}}$ two immersions $\in G$

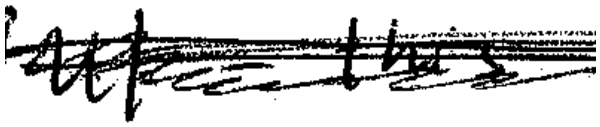
Let us consider categ. of sets U_α and their finite intersections

~~let ... this cat. of sets~~



Original source details: GELFAND-P000578

Uncertain reading. The crossed-out line shows "let", tentatively "this", and "cat. of sets".



Original source details: GELFAND-P000578

Undeciphered passage. The strokes after "let" may include "us". The full passage is indistinct.

GELFAND-P000579

21/X̄-74. Gelfand Sem. Kushnirenko.

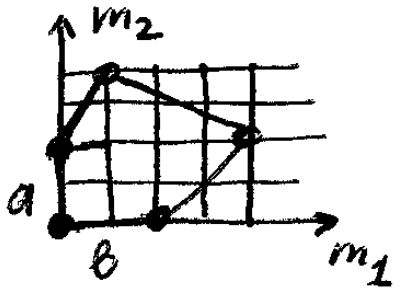
Melvin Hochster.

$P\colon \mathbb{C}^n \to \mathbb{C}$, P — a polynomial, $P = \sum a_m z^m$.

Singular points: $\{z : \frac{\partial P}{\partial z_i} = 0\}$. Consider all the sing. points. Suppose they are isolated.

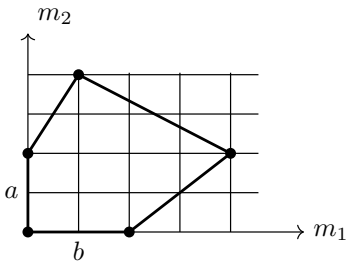
How do we find them, counting multiplicity?

Consider the convex hull of 0 and all $m \in \mathbb{Z}^n$ for which $a_m \neq 0$.



Original source details: GELFAND-P000579

Diagram and typed description.



$$2S - a - b + 1.$$

1) the number of isol. sing. points, counting mult., $\leq 2S - a - b + 1$.

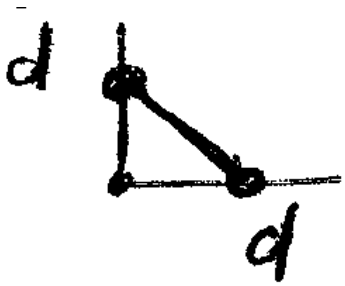
2) if the nondegeneracy cond. holds $= 2S - a - b + 1$.

Idea: consider the degree of the map $S^{2n-1} \to S^{2n-1}$.

$$\underset{z}{|z|=R} \longmapsto \frac{\left(\frac{\partial P}{\partial z_1}, \dots, \frac{\partial P}{\partial z_n}\right)}{|\text{grad } P|}.$$

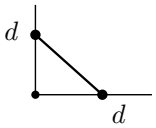
This was known in the homogeneous or quasihomog. case.

$\sum$ of Milnor numbers = degree (abs. fact)



Original source details: GELFAND-P000579

Diagram and typed description.

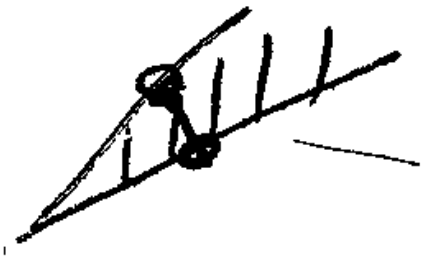


Hochster: consider the ring of monomials with exponents m in the angle.



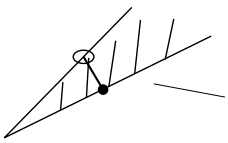
Original source details: GELFAND-P000579

Uncertain reading. The handwritten Russian preposition “в” before “угле” is retraced.



Original source details: GELFAND-P000579

Diagram and typed description.



This is a Cohen–Macaulay ring.

Mumford — toric embeddings.

Polyakov:

Yang–Mills field.

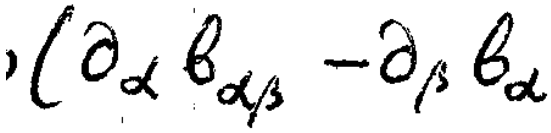
$$S \dots = \int d^4x \left\{ \frac{1}{e^2} \operatorname{Sp}(\partial_\alpha b_{\alpha\beta} - \partial_\beta b_\alpha - [b_\alpha, b_\beta]^2) \right.$$

⟵ action
– Lagrangian
(under the integral sign)



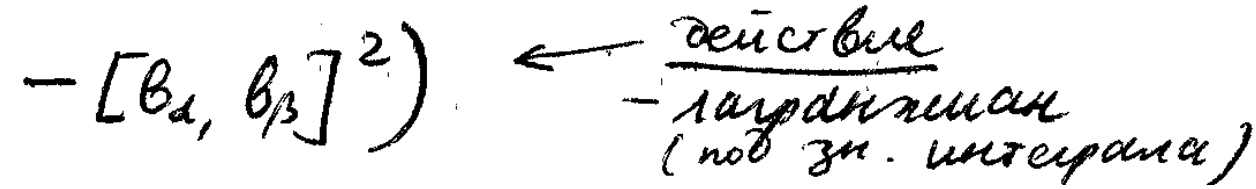
Uncertain reading. A group of marks after S is crossed out. Its reading is obscured by dense strokes.

Original source details: GELFAND-P000579



Uncertain reading. Two subscripts, $\alpha\beta$, are visible after b in the first term.

Original source details: GELFAND-P000579



Original source details: GELFAND-P000579

Uncertain reading. The exponent 2 immediately follows the square bracket. The brace after d^4x is left open.

b_α — elem. of a Lie algebra, given at any point.

There is a vect. bundle over $\mathbb{R}^4$ with fibre a Lie algebra (preferably $SU(2)$)

If $(b_\alpha)^b_a$ — the matrix Γ_α, we obtain $R^b_{\alpha\beta a}$ under the integral sign.

Elem. of a Lie alg. at $\forall$ point $\longleftrightarrow$ connection.

Pontryagin class: $R^\beta_{\alpha,\delta_1\delta_2} \cdot R^\alpha_{\beta,\delta_3\delta_4}$.



Uncertain reading. The two pairs of lower indices read $\delta_1\delta_2$ and $\delta_3\delta_4$.

Original source details: GELFAND-P000579

GELFAND-P000580

28/X-74. Gelfand Sem.

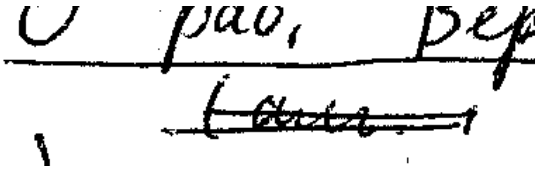
M. I. Graev. On work by Vershik, Graev and Gelfand



Uncertain reading. The second initial has been heavily retraced. Cyrillic “И.” is probable, with “Н.” also possible.

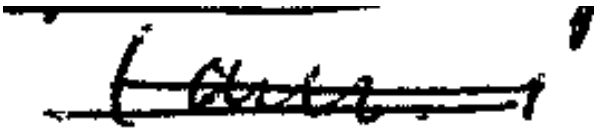
Original source details: GELFAND-P000580

~~(Alg.?)~~



Uncertain reading. The cancelled parenthetical abbreviation below the heading tentatively reads “Алг.” (“Alg.”).

Original source details: GELFAND-P000580



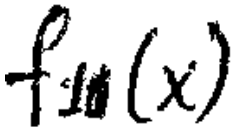
Undeciphered passage. The crossing strokes obscure some of the letters in the abbreviation.

Original source details: GELFAND-P000580

On current algebra for physicists.

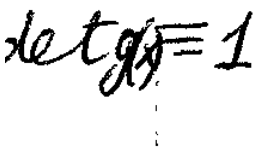
(Funct. Anal., vol. 2, $N = 1$)

$$g(x) = \begin{pmatrix} f_{11}(x) & f_{12}(x) \\ f_{21}(x) & f_{22}(x) \end{pmatrix}, \qquad x \in [0,1], \qquad \det g(x) = 1$$



Uncertain reading. The upper-left entry tentatively has subscript 11. The second digit is overwritten.

Original source details: GELFAND-P000580



Uncertain reading. The strokes at the argument of g overlap. The probable reading is $\det g(x) = 1$.

Original source details: GELFAND-P000580

$f_{ij} : [0,1] \to \mathbb{R}$ continuous functions.

A method for constructing irreducible representations is given, analogous to taking an integral.



Uncertain reading. The line supports reading “взятию” (“taking”). Several final strokes run together.

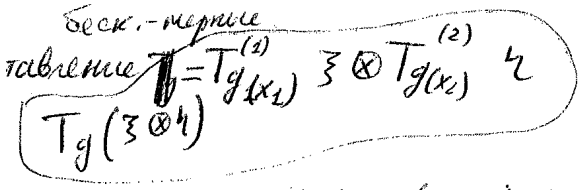
Original source details: GELFAND-P000580

GELFAND-P000581

G. Replace the interval by a set of points. Suppose there are 2 points

$$\begin{matrix} x_1 & x_2 \\ g_1 & g_2 \\ H & \end{matrix} \qquad T^{(1)}_{g_1}\xi \quad T^{(2)}_{g_2}\xi$$

Consider $H_1 \otimes H_2$ and the representation



Original source details: GELFAND-P000581

Diagram and typed description.

infinite-dim.

$T = T^{(1)}_{g(x_1)}\xi \otimes T^{(2)}_{g(x_2)}\eta$

$T_g(\xi \otimes \eta)$



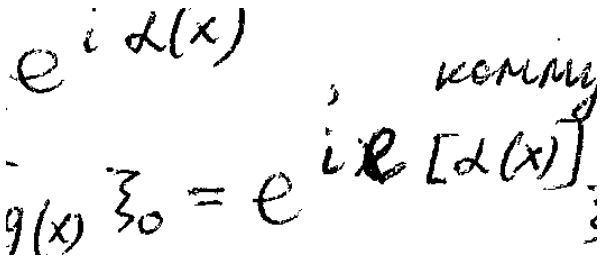
Original source details: GELFAND-P000581

Uncertain reading. The sign before the equals sign resembles T, with heavily retraced lower strokes. T_g is possible. Both operators in the upper pair act on ξ.

Example: $e^{i\alpha(x)}$, commut. group. Rep. (irred.) is 1-dimensional

$$T_{g(x)}\xi_0 = e^{i\ell[\alpha(x)]}\xi_0, \qquad \ell \text{ — lin. functional} \qquad \int \alpha(x)\,dx$$

$\alpha \in C^\infty \Rightarrow \ell$ — a generalized function



Original source details: GELFAND-P000581

Uncertain reading. The letter in the exponent resembles α in the lower formula. A handwritten L is also possible.



Original source details: GELFAND-P000581

Uncertain reading. The Russian abbreviation “непр.” is tentatively expanded as “irreducible”.

matrix

Reps. of groups of order 1 $\Longleftrightarrow$ generalized functions

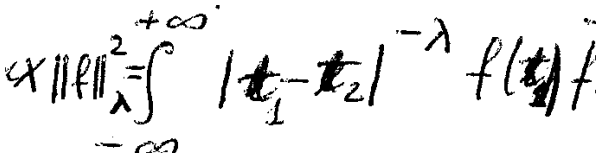
Is there an irred. rep. that is structured in the same way everywhere.

Grp. $SL_2(\mathbb{R})$ representations T_λ in spaces H_λ

H_λ — those $f(t)$ for which

$$\|f\|_\lambda^2 = \int_{-\infty}^{+\infty} |t_1 - t_2|^{-\lambda} f(t_1)\overline{f(t_2)}\, dt_1\, dt_2 < \infty.$$

$0 < \lambda < 1$ for pos. definiteness

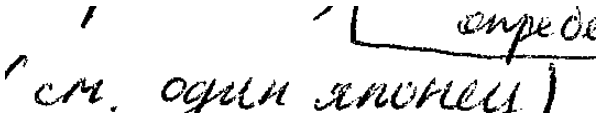


Uncertain reading. One integral sign is visible with two differentials $dt_1 dt_2$. The variable strokes are heavily retraced.

Original source details: GELFAND-P000581

G. There are representations close to 1 (but infinite-dimensional!!)

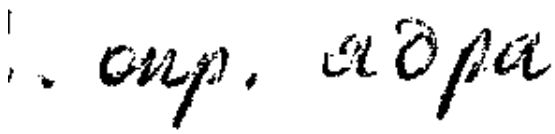
Problem. There is a dimension in Kolmogorov's sense (see one Japanese person).



Uncertain reading. The end of the parenthesis reads "один японец", literally "one Japanese person".

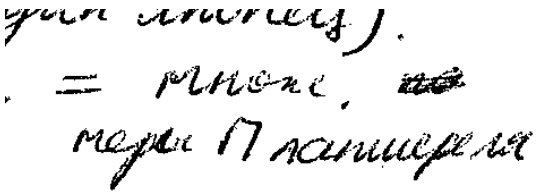
Original source details: GELFAND-P000581

With a proper def. of the kernel we can consider dim in Kolmogorov's sense = mult. ... of the Plancherel measure.



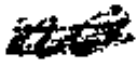
Uncertain reading. The word after the abbreviation for "definition" reads "ядра" ("of the kernel").

Original source details: GELFAND-P000581



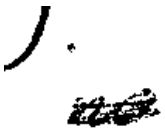
Uncertain reading. The abbreviation after the equals sign tentatively reads "множ." ("mult."). The final strokes overlap.

Original source details: GELFAND-P000581



Uncertain reading. A short group of strokes to the right of the abbreviation is crossed out.

Original source details: GELFAND-P000581



Undeciphered passage. The entire crossed-out group remains unread.

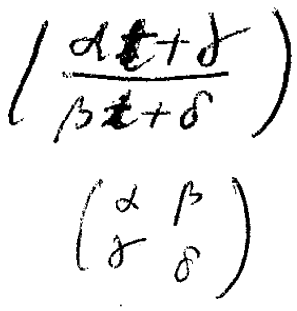
Original source details: GELFAND-P000581

Rep. of $SL_2(\mathbb{R})$,

$$f(t) \longrightarrow f\left(\frac{\alpha t+\gamma}{\beta t+\delta}\right)|\beta t+\delta|^{\rho-1}, \qquad \begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}$$

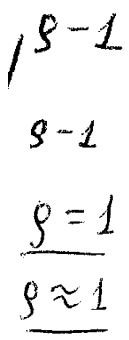
$(\dots)^{\rho-1}$

$\underline{\rho=1}$ — identity rep.
$\underline{\rho\approx 1}$ — close to the identity



Uncertain reading. The looped numerator letter with a rightward stroke matches the lower-left matrix entry, γ. The reading δ remains possible.

Original source details: GELFAND-P000581



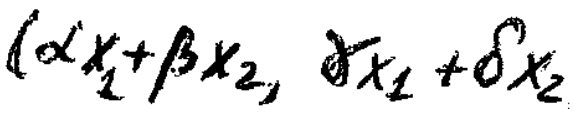
Uncertain reading. The looped sign in both exponents and both following lines is rendered as ρ. s is possible.

Original source details: GELFAND-P000581

$$f(x_1, x_2) \longrightarrow f(\alpha x_1+\beta x_2, \gamma x_1+\delta x_2)$$

on homog. polynomials

$$x_2^n f(x_1/x_2, 1) \qquad \begin{matrix} n \text{ — weight} \\ \approx \text{ dim.}\end{matrix}$$



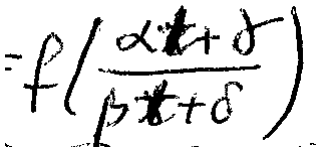
Uncertain reading. Before x_1 in the second argument, the loop and rightward stroke resemble the lower-left matrix entry. The preferred reading is γ, with δ also possible.

Original source details: GELFAND-P000581

Gr.

$$(T_\lambda^{(g)} f)(t) = f\left(\frac{\alpha t+\gamma}{\beta t+\delta}\right)|\beta t+\delta|^{\lambda-2}, \qquad g=\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}$$

$0<\lambda<1$



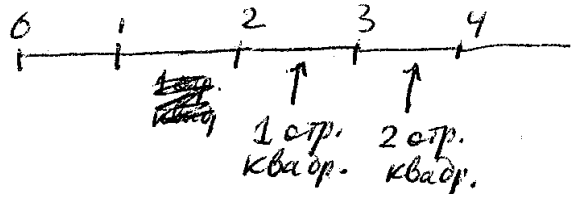
Original source details: GELFAND-P000581

Uncertain reading. The numerator letter resembles the matrix's γ. It is distinguishable from the shape of δ in the denominator, with some uncertainty.

$\lambda=0 \Rightarrow$ the inner prod. degenerates to

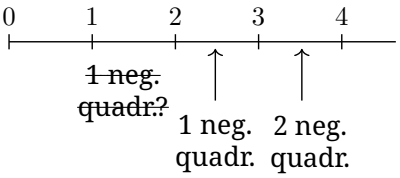
$$\left|\int_{-\infty}^{\infty} f(x)\,dx\right|^2 = \|f\|^2.$$

Analyt. cont.:



Original source details: GELFAND-P000581

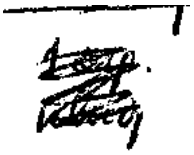
Diagram and typed description.





Original source details: GELFAND-P000581

Uncertain reading. A two-line inscription below the interval 1–2 is crossed out. It tentatively reads "1 neg. quadr."



Original source details: GELFAND-P000581

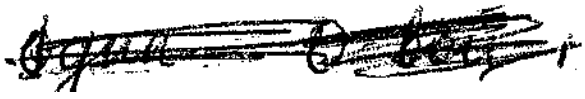
Undeciphered passage. Both lines are only partially legible beneath the cancellation.

$\lambda=0$ — ~~func.? ...~~ obtain ~~...~~ a rep. in a subspace



Original source details: GELFAND-P000581

Uncertain reading. The crossed-out passage after the equality begins with what appears to be "функц." ("func.").



Original source details: GELFAND-P000581

Undeciphered passage. The remaining crossed-out strokes in the phrase are unresolved.



Uncertain reading. A further word is crossed out between “obtain” and “rep.”

Original source details: GELFAND-P000581

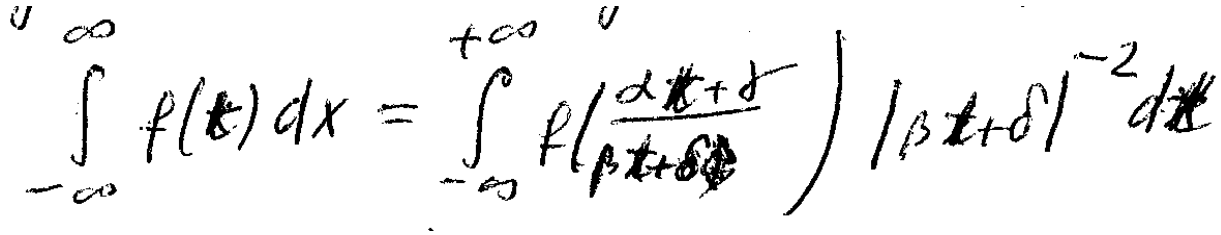


Undeciphered passage. The crossed-out word remains unread.

Original source details: GELFAND-P000581

$$\int_{-\infty}^{\infty} f(t)\,dx = \int_{-\infty}^{+\infty} f\left(\frac{\alpha t+\gamma}{\beta t+\delta}\right)|\beta t+\delta|^{-2}\,dt.$$

($\int$ — invariant)



Original source details: GELFAND-P000581

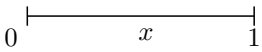
Uncertain reading. The left side has $f(t)\,dx$ and the right has dt. The looped numerator letter is read as γ. Heavy strokes near the denominator’s δ obscure the earlier writing.

GELFAND-P000582

Fix λ, $(0 < \lambda < 1)$ and consider const. matrices $g(x) \equiv \text{const} = g$.



Diagram and typed description.



Original source details: GELFAND-P000582

This is $SL_2(\mathbb{R}) = G$.

Consider the rep. T_λ of this group in H_λ.

Divide the interval into 2 equal parts and take functions const. on the left and right halves

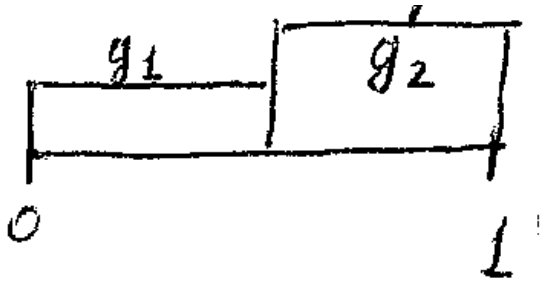
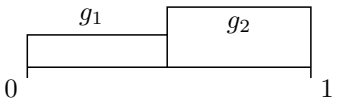


Diagram and typed description.



Original source details: GELFAND-P000582

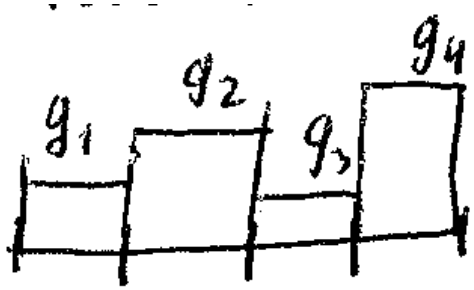
$\overline{\cdots}$ This is $G \times G$.



Original source details: GELFAND-P000582

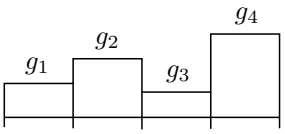
Uncertain reading. A sign before "This is" is heavily crossed out. Its reading is unresolved.

Consider $H_{\lambda/2} \otimes H_{\lambda/2}$ and define there a rep. of $G \times G$



Original source details: GELFAND-P000582

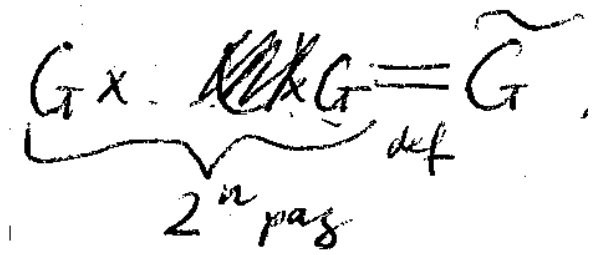
Diagram and typed description.



$$G \times G \times G \times G \longrightarrow \text{rep. in } H_{\lambda/4} \otimes H_{\lambda/4} \otimes H_{\lambda/4} \otimes H_{\lambda/4}.$$

and so on

$$G \subset G \times G \subset G \times G \times G \times G \subset \dots \qquad \varinjlim_{n \to +\infty} \underbrace{G \times \overline{\cdots} \times G}_{2^n \text{ times}} \underset{\text{def}}{=} \widetilde{G}.$$



Original source details: GELFAND-P000582

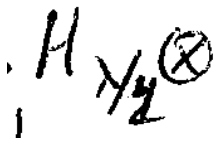
Uncertain reading. A group of symbols between the outer factors is crossed out. The brace reads "2^n times".

Representation of $\widetilde{G}$?

We need to embed $H_\lambda \subset H_{\lambda/2} \otimes H_{\lambda/2}$ compatibly with the group action.

Decomp. into irreducibles — in the discrete part enters H_λ once, and in the complement, the rest, and so on. This lets us embed

$$H_\lambda \hookrightarrow H_{\lambda/2} \otimes H_{\lambda/2} \hookrightarrow \underbrace{H_{\lambda/4} \otimes \dots \otimes H_{\lambda/4}}_{4 \text{ times}} \hookrightarrow .$$



Original source details: GELFAND-P000582

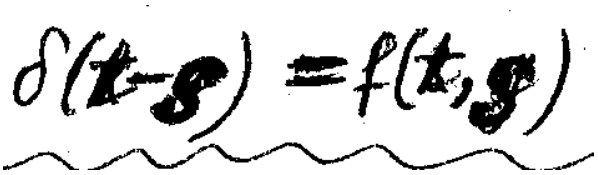
Uncertain reading. The denominator of the first subscript under the brace is overwritten. The probable reading is $\lambda/4$.

$$H_{\lambda/2} \otimes H_{\lambda/2} = H_\lambda \oplus \text{the rest}.$$

The embedding of H_λ is determined (up to a factor)

$$H_\lambda \hookrightarrow H_{\lambda/2} \otimes H_{\lambda/2}$$

$$f(t) \longrightarrow f(t)\delta(t-s) = f(t,s) \qquad \text{(unit. embedding)}$$



Uncertain reading. Both occurrences of the second argument are overwritten with a looped sign. s is a tentative reading.

Original source details: GELFAND-P000582



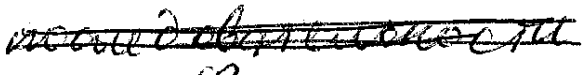
Uncertain reading. The last letter of "вложение" ("embedding") is heavily retraced. The ending of "вложения" may underlie it.

Original source details: GELFAND-P000582

$$\varinjlim H_{\lambda/2^n} \otimes \ldots \otimes H_{\lambda/2^n}$$ and take the completion

the group of step functions acts there. The rep. extends to a rep. of the group of cont. functions.

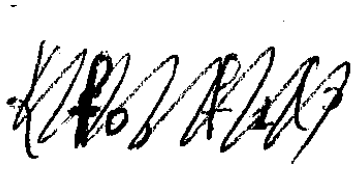
Elements of the space — ~~sequences?~~ functions $f(x,t)$, $-\infty < t < +\infty$, $0 < x < 1$



Uncertain reading. The crossed-out word resembles "последовательности" ("sequences"). Some letters are covered.

Original source details: GELFAND-P000582

~~$f(t_0)f(t)$~~? $\int_{-\infty}^{\infty} f(x,t)\,dt = 1$



Uncertain reading. The cancellation to the left of the integral resembles $f(t_0)f(t)$. The complete reading is unresolved.

Original source details: GELFAND-P000582

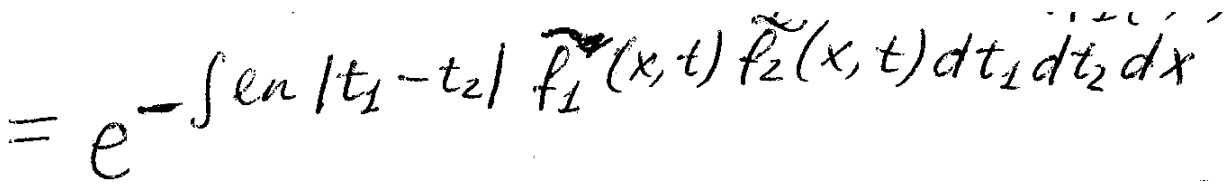
Consider formal lin. combs. $\sum \lambda_i \cdot f_i(x,t)$.

$\sum \lambda_i \cdot 0 \neq 0$ (!).

Inner prod.

$$\langle f_1(x,t), f_2(x,t)\rangle = e^{-\int \ln|t_1-t_2|\, \widetilde{f}_1(x,t)\widetilde{f}_2(x,t)\,dt_1\,dt_2\,dx}$$

and by linearity



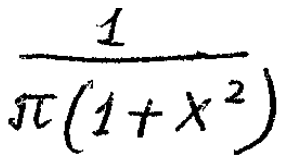
Original source details: GELFAND-P000582

Uncertain reading. One integral sign is written, and both arguments of $\widetilde{f}_i$ are written (x,t). The tilde over the first f is retraced.

$f_1(x,t)$ — a curve in a Hilb. space. Inner prod. of δ-functions on them.

GELFAND-P000583

$$\widetilde{f}(x,t) = f(x,t) - \frac{1}{\pi(1+x^2)}.$$

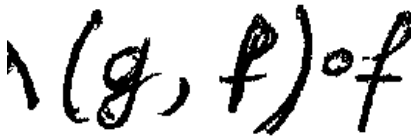


Uncertain reading. The denominator of the subtracted fraction contains x^2.

Original source details: GELFAND-P000583

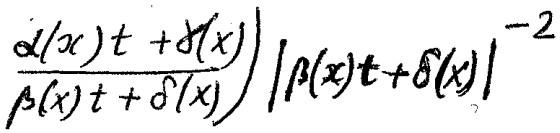
Representation:

$$U(g(x))f(x,t) = \lambda(g,f) \circ f\left(x, \frac{\alpha(x)t+\gamma(x)}{\beta(x)t+\delta(x)}\right)|\beta(x)t+\delta(x)|^{-2}$$



Uncertain reading. A small open circle is visible between $\lambda(g,f)$ and f.

Original source details: GELFAND-P000583



Uncertain reading. The numerator has a loop with a long rightward stroke, matching γ on frame 581. The denominator's δ has an upper hook.

Original source details: GELFAND-P000583

GELFAND-P000584

17/II-75 Gelfand Sem.
Tulovskii FIO — I and II.

$$\Omega \subset \mathbb{R}^n, \qquad u \in \mathcal{D}'(\Omega), \qquad \mathrm{WF}(u) \subseteq \Omega \times (\mathbb{R}^n \setminus 0).$$

~~$u \in \mathcal{D}'(\Omega)$~~



Uncertain reading. The cancelled formula resembles $u \in \mathcal{D}'(\Omega)$.

Original source details: GELFAND-P000584

Definition. $(x_0, \xi_0) \notin \mathrm{WF}(u)$, if $\exists\, \varphi(x) \in C_0^\infty(\Omega)$, $\varphi(x_0) \neq 0$

$$|\widetilde{u\varphi}(\xi)| \leq C_N(1+|\xi|)^{-N} \quad \text{when} \quad |\dot{\xi} - \dot{\xi}_0| < \varepsilon \qquad \left(\dot{\xi} = \frac{\xi}{|\xi|}\right)$$

G.



Original source details: GELFAND-P000584

Diagram and typed description.



Crossed-out diagram. The cancellation partly covers the upper arcs.

Smoothness of a function of one var., obtained as an $\int$ over parallel hyperplanes — $\sim$ membership of the covector in the wavefront.

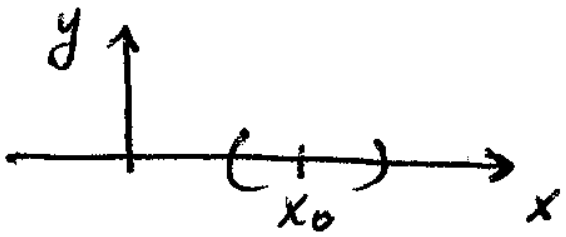


Original source details: GELFAND-P000584

Uncertain reading. The initial letter of “фронту” (“front”) is heavily overwritten. The earlier layer is unresolved.

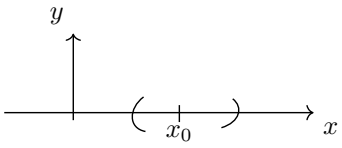
— · —

1 variable.



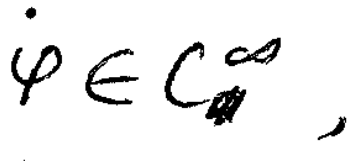
Original source details: GELFAND-P000584

Diagram and typed description.



$$(x_0, \xi_0) \notin \mathrm{WF}(u), \quad (\xi_0 > 0), \quad \text{if} \quad u = \varphi(x) + u_1 \quad \text{(locally)}$$

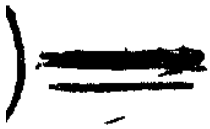
$\varphi \in C_0^\infty$, u_1 analytically continues into the upper half-plane.



Original source details: GELFAND-P000584

Uncertain reading. The overwritten subscript in C_0^∞ resembles 0. A separate dot is visible above φ.

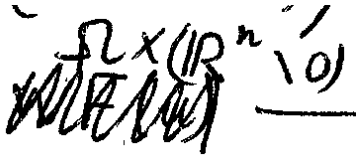
$$\pi(\mathrm{WF}(u)) = \operatorname{sing\,supp} u. \qquad (\pi : T^*\Omega \to \Omega).$$



Uncertain reading. The lines read as $=$ overlap a heavier, arrow-like stroke.

Original source details: GELFAND-P000584

$$\pi' : \overset{\Omega\times(\mathbb{R}^n\setminus 0)}{\cancel{\mathrm{WF}(u)}} \longrightarrow \mathbb{R}^n \setminus 0.$$



Uncertain reading. The domain $\mathrm{WF}(u)$ is crossed out. $\Omega \times (\mathbb{R}^n \setminus 0)$ is inserted above it.

Original source details: GELFAND-P000584

$\pi'\mathrm{WF}(u)$ — the smallest closed cone outside which the Fourier transform decays rapidly.



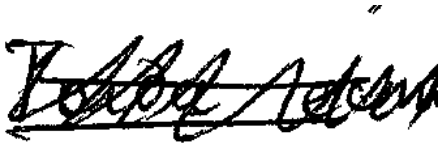
Uncertain reading. The initial letter of “Фурье” (“Fourier”) is retraced, with short strokes above and below.

Original source details: GELFAND-P000584

Let $\Gamma \subseteq \Omega \times (\mathbb{R}^n \setminus 0)$ — an open conic set,

$$\begin{aligned}
&\Gamma_1, \Gamma_2 \subseteq \Omega \times (\mathbb{R}^n \setminus 0),\\
&\Gamma_1 + \Gamma_2 = \{(x; \xi + \eta) : (x, \xi) \in \Gamma_1,\ (x, \eta) \in \Gamma_2\},\\
&\mathcal{D}'_\Gamma(\Omega) = \{u \in \mathcal{D}'(\Omega) : \mathrm{WF}(u) \subseteq \Gamma\}.
\end{aligned}$$

~~Then ...~~ let $\Gamma_1 + \Gamma_2 \subseteq \Omega \times (\mathbb{R}^n \setminus 0)$.



Undeciphered passage. Before “let”, “Then” and a further word are crossed out. The second word remains partly unread.

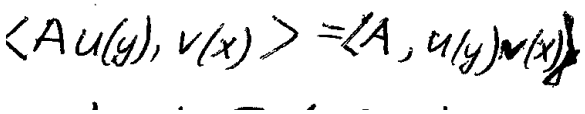
Original source details: GELFAND-P000584

Then $\exists$ a bilin. map:

$$\mathcal{D}'_{\Gamma_1}(\Omega) \times \mathcal{D}'_{\Gamma_2}(\Omega) \longrightarrow \mathcal{D}'_{\Gamma_1+\Gamma_2\cup\Gamma_1\cup\Gamma_2}(\Omega).$$

—— · ——

$$\begin{aligned}
&\Omega_x \subseteq \mathbb{R}^{n_x},\\
&\Omega_y \subseteq \mathbb{R}^{n_y}, \qquad A \in \mathcal{D}'(\Omega_x \times \Omega_y).\\
A : C_0^\infty(\Omega_y) \to \mathcal{D}'(\Omega_x), \qquad &\langle Au(y), v(x)\rangle = \langle A, u(y)v(x)\rangle
\end{aligned}$$



Uncertain reading. The strokes near v and the closing sign on the right are heavily retraced.

Original source details: GELFAND-P000584

(x, ξ, y, η) — $\mathrm{WF}(\text{of the operator}) \underset{\text{def}}{=} \mathrm{WF}(\text{of the kernel})$

Thm. Suppose for simplicity that in $\mathrm{WF}(A)$ there are no points where $\xi = 0$ or $\eta = 0$
Then $A : C_0^\infty(\Omega_y) \to C^\infty(\Omega_x)$ and extends to a map $A : \mathcal{E}'(\Omega_y) \to \mathcal{D}'(\Omega_x)$.
Moreover: if $u \in \mathcal{E}'(\Omega_y)$, then $\mathrm{WF}(Au) \subseteq \mathrm{WF}'(A) \circ \mathrm{WF}(u)$,

GELFAND-P000585

where

$$\mathrm{WF}'(A) = \{(x, \xi, y, -\eta), \text{ where } (x, \xi, y, \eta) \in \mathrm{WF}(A)\}.$$

$$(x, \xi) \in \mathrm{WF}'(A) \circ \mathrm{WF}(u) \iff \exists\, (x, \xi, y, \eta) \in \mathrm{WF}'(A), \quad (y, \eta) \in \mathrm{WF}(u).$$

— · —

G.'s idea — integrate both lin. functions and polynomials.



Original source details: GELFAND-P000585

Diagram and typed description.



already different.



Original source details: GELFAND-P000585

Uncertain reading. The ending favors Russian "другое". "Другая" is a less likely reading.

— · —

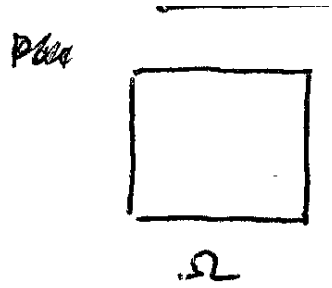
$$\frac{A : \mathcal{E}'(\Omega_y) \to \mathcal{D}'(\Omega_x) \qquad B : \mathcal{E}'(\Omega_z) \to \mathcal{E}'(\Omega_y)}{\mathrm{WF}'(A \circ B) \subseteq \mathrm{WF}'(A) \circ \mathrm{WF}'(B)}$$



Original source details: GELFAND-P000585

Uncertain reading. Both occurrences of WF on the right of the inclusion carry a prime.

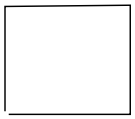
Sinai. Stat. physics (from the end)
Gibbs random field.



Original source details: GELFAND-P000585

Diagram and typed description.

Fig.?



Ω



Original source details: GELFAND-P000585

Uncertain reading. Crossed-out vertical strokes lie to the left of the outline. “II” and “III” are possible.



Original source details: GELFAND-P000585

Uncertain reading. The upper-left label resembles Russian “Рис” (“Fig.”). $\mathbb{R}^n$ is also possible.

G. Consider a measure μ on countable subsets of $\mathbb{R}^n$ without accumulation points (configurations denoted by X).

Measurable sets — the smallest σ-algebra generated by cylinder sets:

$$\{K : |X \cap K| = n\} = \text{Ц}_{K,n} \qquad (K \text{ — compact})$$

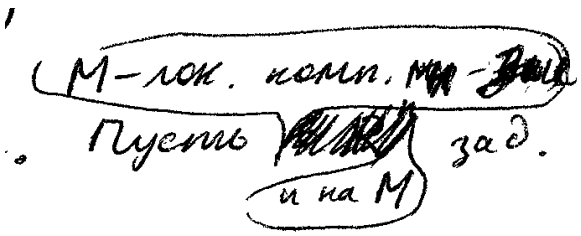
The measure is countably additive.

Main example: Poisson distribution.

(M — loc. comp. ...)
Let ... be given

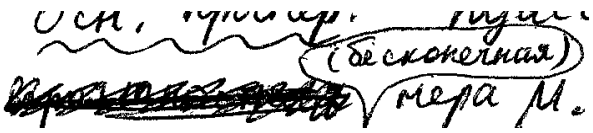
and on M

(infinite)
... measure μ. Then



Undeciphered passage. Strokes overlap at the end of the upper insertion and in the notation after “Let”. The lower insertion reads “and on M”.

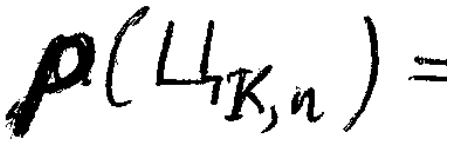
Original source details: GELFAND-P000585



Original source details: GELFAND-P000585

Undeciphered passage. A phrase before “measure” is heavily crossed out. “Infinite” is inserted in an outline above it.

$$\rho(\text{Ц}_{K,n}) = \frac{m(K)^n}{n!} e^{-m(K)}$$



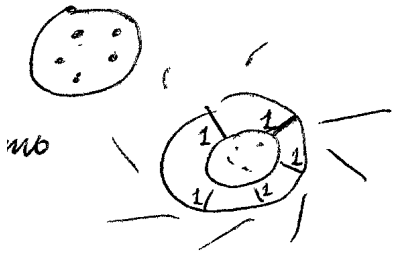
Original source details: GELFAND-P000585

Uncertain reading. The first symbol is heavily retraced and resembles ρ. The numerator and exponent use m.

— · —

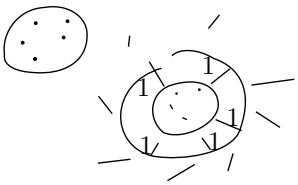
Gibbs distributions:

“Interaction is finite”: independence.



Original source details: GELFAND-P000585

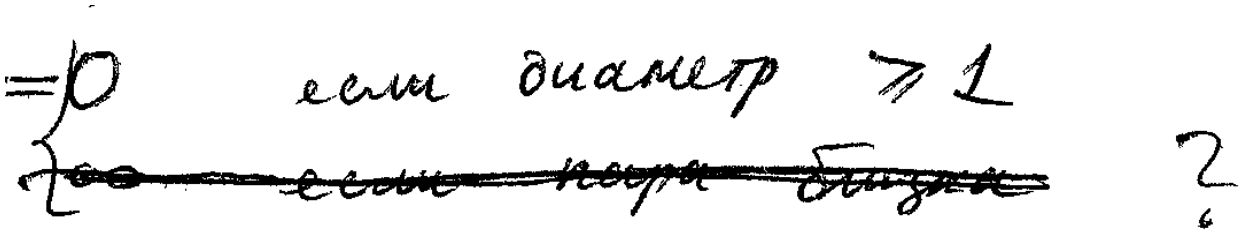
Diagram and typed description.



The upper contour encloses five dots. Some of the five segments in the lower drawing stop short of the inner contour.

Consider $U(x_1, x_2)$, $U(x_1, x_2, x_3)$, …

$$U(x_1, \dots, x_n) = \begin{cases} 0 & \text{if the diameter is } \geqslant 1 \\ \infty \text{ if} & \text{... close?} \quad ? \end{cases}$$



Original source details: GELFAND-P000585

Undeciphered passage. The second case is cancelled. The infinity sign and “if” are visible, followed by what appears to end in “close”. The intervening strokes are obscured.

The seq. breaks off, since it becomes $\equiv +\infty$.



Original source details: GELFAND-P000585

Uncertain reading. The word after “since” reads “становится” (“becomes”).

Consider countable configurations.

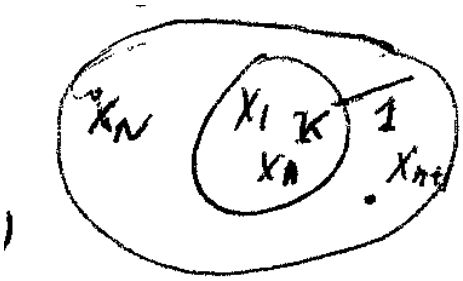
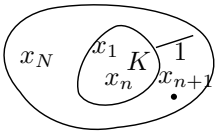


Diagram and typed description.

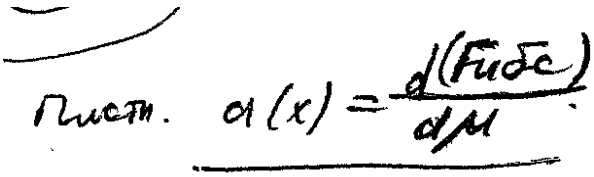


The right subscript tentatively reads $n+1$. The final strokes meet the outline.

Original source details: GELFAND-P000585

$$d(X) = e^{-\sum U_k(x_1,\ldots,x_k)}$$

Density $d(x) = \dfrac{d(\text{Gibbs})}{d\mu}$.



Uncertain reading. The abbreviation to the left of the fraction resembles “Плотн.” (“density”).

Original source details: GELFAND-P000585

GELFAND-P000586

S. Consider $\underbrace{K \times \ldots \times K}_{n \text{ times}}$ with measure $m \times \ldots \times m$.

$$\frac{dG(X)}{dm} = d(X)$$

- -

GELFAND-P000587

24?/II–75. Gelfand’s sem.



Uncertain reading. The second digit of the day is overwritten. The reading 24 is tentative.

Original source details: GELFAND-P000587

Ya. Sinai. Let $X = \{x\}$ be an infinite discrete set in $\mathbb{R}^2$,



Undeciphered passage. A short mark after the comma is crossed out.

Original source details: GELFAND-P000587

m — a measure in $\mathbb{R}^2$. $\mathcal{M}$ — all such sets.

Poisson measure on $\mathcal{M}$: let O be a region.

$$\mu_0(|X \cap O| = k) = e^{-m(O)}[m(O)]^k/k! \qquad (1)$$

$m(O)$ — expected number of particles found in O.

More generally: (M, m) — a space with a measure [inserted below: “and a metric”]
$m(M) < \infty$ — uninteresting case

Interesting case: $m(M) = \infty$ and certain homogeneity properties.



Undeciphered passage. A short mark is crossed out above “certain”.

Original source details: GELFAND-P000587

Let 2 functions be given: $U_1(x)$ [below: “bounded”] and

$$U_2(x, y) = \begin{cases} \infty & \text{if } \operatorname{dist}(x, y) < \varepsilon, \\ 0 & \text{if } \operatorname{dist}(x, y) \geq r_0. \end{cases}$$

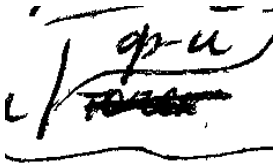


Uncertain reading. The overwritten sign in the second row reads as 0.

Original source details: GELFAND-P000587

($U_2(x, y)$ is symmetric — a function on a pair of points.)

A Gibbs measure is constructed from a pair of [inserted: “functions”] U_1, U_2.



Undeciphered passage. The word beneath the insertion “functions” is crossed out.

Original source details: GELFAND-P000587

Def. by means of conditional probabilities.

Let there be a measure μ on the space of configurations $\mathcal{M}$

$$X = O \cap \ldots, \quad \underline{Y} = O' \cap \ldots$$



Uncertain reading. The overwritten sign before $\cap$ reads as O'.

Original source details: GELFAND-P000587

x — inside, y — outside (in the given configuration)

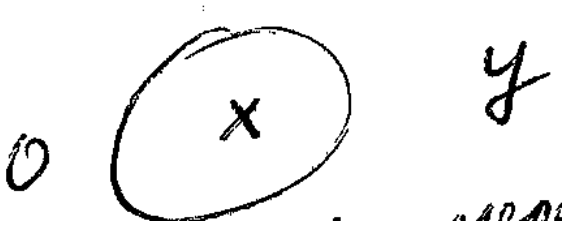
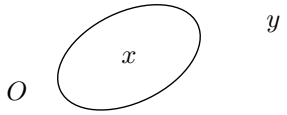


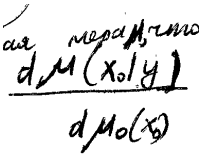
Diagram and typed description.



Original source details: GELFAND-P000587

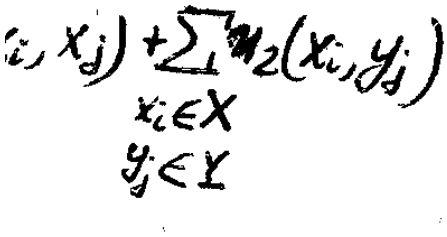
A Gibbs measure is a measure such that:

$$\frac{d\mu(x_0 \mid y)}{d\mu_0(x_0)} = \frac{1}{\Xi(y;O)} e^{-\left[\sum_{x_i \in X} U_1(x_i) + \sum_{(x_i,x_j)\in X} U_2(x_i,x_j) + \sum_{\substack{x_i \in X \\ y_j \in Y}} U_2(x_i,y_j)\right.}$$



Uncertain reading. A subscript 0 is visible after x in the denominator.

Original source details: GELFAND-P000587



Uncertain reading. The first symbol in the final summand is overwritten and tentatively reads U_2. The closing square bracket is not visible.

Original source details: GELFAND-P000587

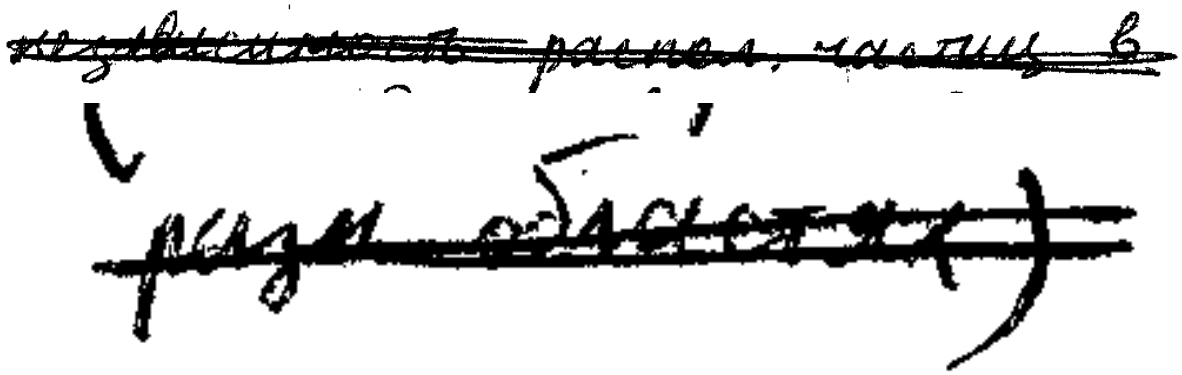
$$x = (x_1, \ldots, x_k) \qquad X = \{X_0 = X \cap O,\ Y = X \setminus X_0\}$$

All this depends only on the r_0-neighbourhood of O (on those points y, that fell within it).

$\mu(x \mid y)$ — a measure on configurations in O.

$\mu_0(x)$ — a measure —"— O.

(In def. (1) it is understood, ~~the independence of different particles in different regions)~~ that we take the σ-algebra generated by the corresponding cylinder sets.)



Original source details: GELFAND-P000587

Uncertain reading. The first word in the crossed-out clause tentatively reads “independence”.

[G. $\mu(x \mid y)$ is defined because there is a measurable partition (Fubini's theorem — general) The measure on Y is induced by the measure on the whole space, by means of preimages.] Let there be 2 regions.

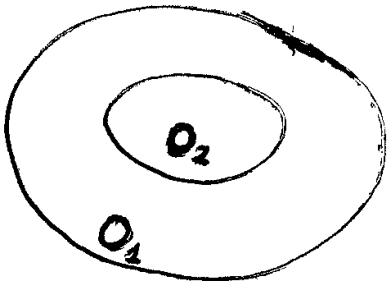
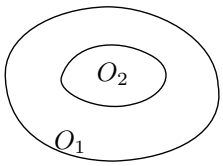


Diagram and typed description.

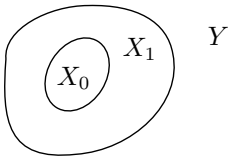


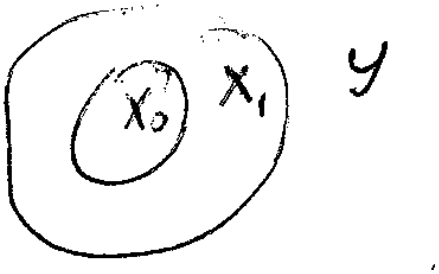
Original source details: GELFAND-P000587

From the measure μ on X a measure is constructed on Y (complete in all possible ways with the set X_0).

GELFAND-P000588

Diagram and typed description.





Original source details: GELFAND-P000588

Fix X_0, X_1 and Y. Then the (cond.) measure is calculated in two ways:

Theorem. ∃ at least one Gibbs distribution.

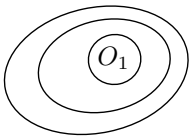
Undeciphered passage. A short mark before ∃ is crossed out.



Original source details: GELFAND-P000588

Proof.

Diagram and typed description.





Original source details: GELFAND-P000588

$O_1 \subset O_2 \subset O_3 \subset \ldots$ and the boundary conditions are fixed.

~~Let~~ The space of config. in O_1 is compact. $\exists$ a subseq. having a limit

and so on ...

Ex. 1 M — a line, m — Lebesgue measure $\Rightarrow$ there is only one Gibbs distribution.

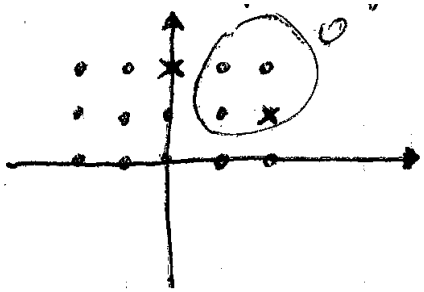
Ex. 2 $M = \mathbb{R}^k$, $U_1 \equiv 0$, $m(O) \leq c\ell(O)$ (ℓ — Lebesgue measure)

For small c the Gibbs distrib. is unique.

— · —

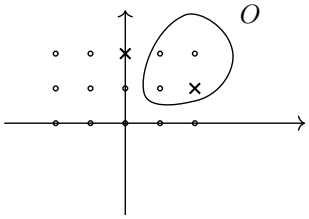
Lattice Gibbs distrib.

Consider the lattice in $M = \mathbb{Z}^d$, $d \geq 2$.



Original source details: GELFAND-P000588

Diagram and typed description.



The lattice has two crosses and round marks. Region O encloses four sites. The drawing leaves the meaning of the marks unlabelled.

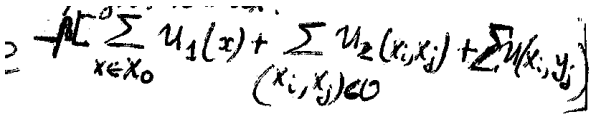
X — infinite subsets of lattice points

$\mathcal{M}$

μ_0 prob. that a particle is present $\frac{1}{2}$, a particle is absent $\frac{1}{2}$, and independence at different sites.

O — a finite region.

$$\mu(X_0 \mid Y) = \frac{1}{\Xi(Y,O)} e^{-\beta\left[\sum_{x\in X_0} U_1(x) + \sum_{(x_i,x_j)\in O} U_2(x_i,x_j) + \sum U(x_i,y_j)\right]}$$



Uncertain reading. No subscript is visible on U in the final sum. The symbol before the bracket reads β.

Original source details: GELFAND-P000588

U_2 also vanishes for large $|x-y|$.

$U_1 \equiv 0$, U_2 — small $\Rightarrow$ one Gibbs distrib.

(perturbation, prob. joint distrib. of independent random variables).

$$U_1(x) = \alpha \qquad U_2(x_i, x_j) = \cancel{U_2(|x_i - x_j|)}\, U_2(x_i - x_j)$$

$$U_2(-x) = U_2(x).$$

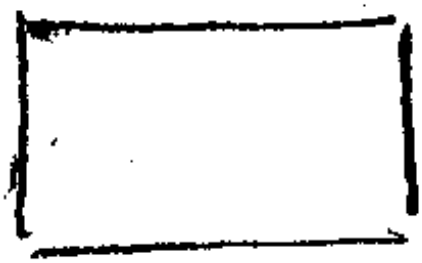


Original source details: GELFAND-P000588

Uncertain reading. A horizontal stroke is visible between x_i and x_j. The vertical strokes and outer parentheses of the cancelled formula are tentative.

1. Periodic ground state.

X — a periodic configuration (repetition of a rectangle)

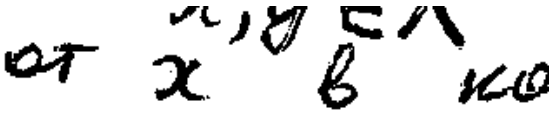


Original source details: GELFAND-P000588

Diagram and typed description.

$$H(X) = \sum_{x \in X} U_1(x) + \sum_{x,y \in X} U_2(x,y) \qquad \text{(formal expression)}$$

Let X' be nonperiodic and differ from x in finitely many places. Write $H(X')$ (also formally).

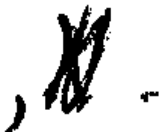


Original source details: GELFAND-P000588

Uncertain reading. The symbol after "from" is a lowercase x.

If $\underbrace{H(X') - H(X) \geq 0}_{\text{already def.}}$ $\forall\, X'$, then X is called a ground state.

1) Let $(X_1, X_2, \cancel{\ldots}, \ldots, X_k) = G(U)$ — the set of periodic ground states corresponding to U.



Original source details: GELFAND-P000588

Undeciphered passage. One mark before the ellipsis is crossed out.

GELFAND-P000589

2) Peierls condition (nondegeneracy of ground states):

(so that ground states — the main contribution to $e^{-\beta H}$).



Undeciphered passage. One mark between the dash and “main” is crossed out.

Original source details: GELFAND-P000589

$$H(X') - H(X) > \dots$$



Undeciphered passage. A short entry before “Let” is heavily crossed out.

Original source details: GELFAND-P000589

Let $\bar{X}$ — an arbitrary configuration.

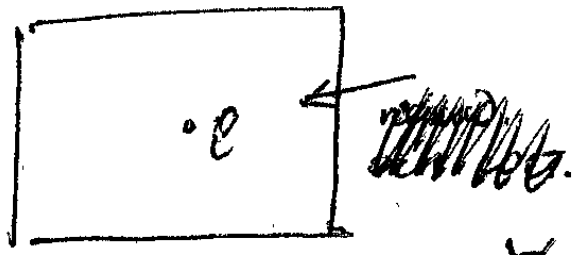
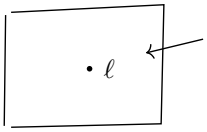


Diagram and typed description.



Original source details: GELFAND-P000589



Undeciphered passage. The label by the arrow is crossed out.

Original source details: GELFAND-P000589

ℓ is called a boundary point of $\bar{X}$, if in a neighbourhood of ℓ $\bar{X}$ does not agree with any of the periodic ground states.

Suppose $\forall$ config. X', differing from X in finitely many points,

Peierls cond.:

$$H(X') - H(X) \geq g\,\Gamma(X') \qquad (g > 0).$$

$\Gamma(X') \longleftarrow$ number of boundary points in X'.



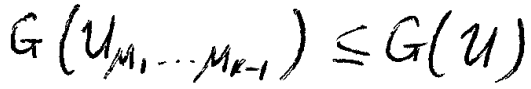
Undeciphered passage. A crossed-out entry stands between g and $\Gamma(X')$.

Original source details: GELFAND-P000589

— · —

Include $\underset{(\text{pair } U_1, U_2)}{U}$ in a $(k-1)$-parameter family $U_{\mu_1,\dots,\mu_{k-1}}$, $U_{0,\dots,0} = U$ (there are k ground states — here the same k).

$$G(U_{\mu_1,\dots,\mu_{k-1}}) \le G(U) \qquad \text{for small } \mu$$



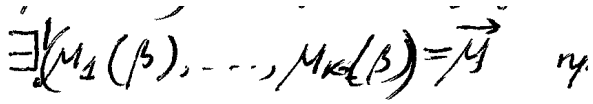
Uncertain reading. The sign between the two G expressions reads as $\le$.

Original source details: GELFAND-P000589

(ground states can only disappear). Let the perturbations be “lin. indep.”.

Then $\exists \beta_0, \forall \beta > \beta_0$

$$\exists!\big(\mu_1(\beta), \dots, \mu_{k-1}(\beta)\big) = \vec{\mu}$$



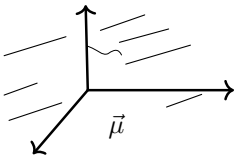
Uncertain reading. An exclamation mark is visible after $\exists$. The last subscript reads as $k-1$, with its end touching the parenthesis.

Original source details: GELFAND-P000589

for which there are k Gibbs distributions. Through it pass k curves, on which there are $(k-1)$ Gibbs distributions, C_k^2 surfaces, on which there are $(k-2)$ Gibbs distributions and so on.



Diagram and typed description.

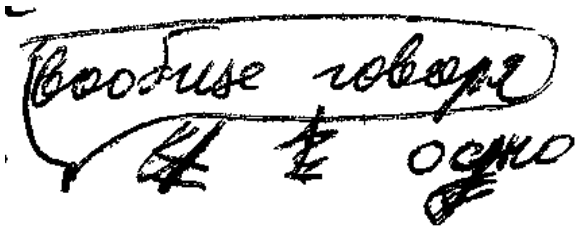


The grouping of the short strokes is approximate.

Original source details: GELFAND-P000589

With shifts one obtains, “as a rule”, one ground state.

Nondegeneracy: as $\mu_1, \dots, \mu_k$ vary there is [inserted: “generally speaking”] one ground state, on the curves $\le (k-1)$ and so on.



Uncertain reading. Two altered signs precede “one”.

Original source details: GELFAND-P000589

Then the same holds for Gibbs distributions.

Gelfand’s question: Gibbs distrib. in the p-adic case.

GELFAND-P000590

[The upper part of the frame contains no text.]

—— · ——

3/III–75 Gelfand's sem.

V. N. Tulovskii. WF. $\Omega \subset \mathbb{R}^n$, $u \in \mathcal{D}'(\Omega)$.

$(x_0, \xi_0) \notin \mathrm{WF}(u)$, if $(x_0, \xi_0) \in \Omega \times (\mathbb{R}^n \setminus 0)$ and from $\phi'(x_0) = \xi_0$ it follows that

$$\left|\langle u, \varphi e^{i\lambda\phi}\rangle\right| \le C_N \lambda^{-N}, \qquad \forall N$$

(the linearity of ϕ does not matter)

—— · ——

Let $\Omega \subset \mathbb{R}^n$, $\theta \in \mathbb{R}^N$, $a(x,\theta) \in S^m_{1,0}(\Omega \times \mathbb{R}^N)$.

Φ — phase function.

$$\langle A, u\rangle = \int a(x,\theta) e^{i\phi(x,\theta)} u(x)\, dx\, d\theta =$$

GELFAND-P000591

$$= \lim_{\varepsilon \to 0} \int a(x,\theta) e^{i\phi(x,\theta)} u(x) \chi(\varepsilon\theta)\, dx\, d\theta$$

Φ - фазовая ф-я $\quad \langle A, u\rangle = \int a(x,\theta)\, e^{i\,\varphi(x,\theta)}\, u(x)\, dx\, d\theta =$

Original source details: GELFAND-P000590

Uncertain reading. An uppercase Φ precedes "phase", and a lowercase ϕ appears in the exponent.



Uncertain reading. The overwritten sign before (x) tentatively reads u.

Original source details: GELFAND-P000590

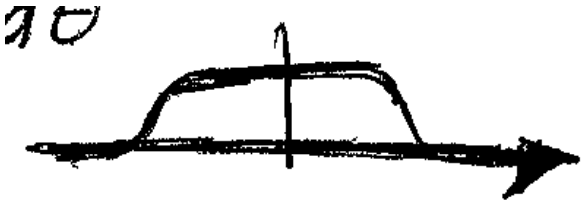
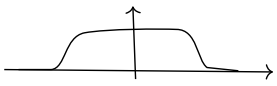


Diagram and typed description.

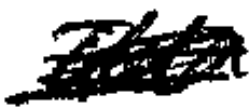


Original source details: GELFAND-P000591

Let $\phi'_\theta(x_0, \theta_0) \neq 0$. Write the decomposition $a = a_1 + a_2$,

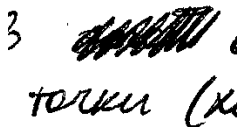
(example: $a = 1$, $\phi = \langle x, \theta\rangle$, $\langle A, u\rangle = u(0)$, i.e. $A(x) = \delta(x)$.).

Undeciphered passage. A short entry before "i.e." is crossed out.



Original source details: GELFAND-P000591

where a_2 is supported in a conic neighbourhood of (x_0, θ_0), and $a_1(x, \theta) = 0$ in a neighbourhood of the point (x_0, θ_0).



Original source details: GELFAND-P000591

Undeciphered passage. An entry after "in" is crossed out. "Point" is inserted below the line.

$$\langle A_2, u\rangle = \int a_2(x,\theta) e^{i\phi(x,\theta)} u(x)\, dx\, d\theta, \qquad \text{here } \phi'_\theta \neq 0 \Rightarrow A_2 \in C^\infty.$$

$$C_\phi = \{(x,\theta) : \phi'_\theta = 0\}$$

Theorem. ~~WF~~ Consider the map

$$\lambda_\phi : C_\phi \longrightarrow \Omega \times (\mathbb{R}^n \setminus 0),$$
$$(x,\theta) \longrightarrow (x, \phi'_x).$$

Denote $\lambda_\phi(C_\phi) = \Lambda_\Phi$.

Then $\mathrm{WF}(A) \subset \Lambda_\Phi$.



Original source details: GELFAND-P000591

Uncertain reading. The sign following WF(A) is retraced and reads as $\subset$.

—— · ——

Nondegenerate phase functions — if $\mathrm{rank}(\phi''_{\theta x}, \phi''_{\theta\theta}) = N$ on C_ϕ.

Then λ_ϕ — a local diffeom. and Λ_ϕ — a conic Lagrangian manifold, i.e. $\langle \xi, dx\rangle|_{\Lambda_\phi} = 0$.



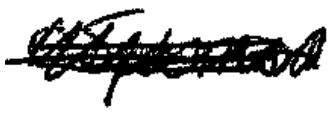
Original source details: GELFAND-P000591

Undeciphered passage. A short mark between "and" and Λ_ϕ is crossed out.

Conversely: $\forall$ conic Lagr. manifold can locally be obtained from a nondegenerate phase function (and one can take $N = n$).

—— · ——

~~Let there be 2 nondegenerate phase functions.~~



Original source details: GELFAND-P000591

Undeciphered passage. The crossed-out ending tentatively begins "Обра…" ("Converse…").

Check of the Lagrangian property:

$$\phi'_\theta(x,\theta) = 0 \;\Rightarrow\; \phi(x,\theta) = 0$$

$$\phi'_\theta(x+dx,\theta+d\theta) = 0, \qquad \phi'_x\,dx + \underbrace{\phi'_\theta\,d\theta}_{\text{“0”}} = 0.$$

hence $\phi'_x\,dx = 0$ on Λ_ϕ, or $\xi\,dx = 0$ on Λ_ϕ. ▷.

—— · ——

Let there be 2 phase functions ϕ_1 and ϕ_2,

$$\lambda_{\phi_1} : C_{\phi_1} \to \Lambda_\phi, \qquad \lambda_{\phi_2} : C_{\phi_2} \to \Lambda$$



Uncertain reading. A subscript is visible on the first target Λ and tentatively reads ϕ.

Original source details: GELFAND-P000591

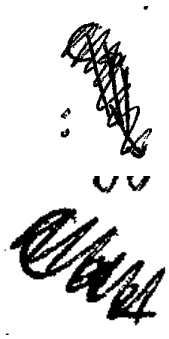
(Λ is the same).

$$(x_0,\theta_1^0) \longrightarrow (x_0,\xi_0),$$
$$(x_0,\theta_2^0) \longrightarrow (x_0,\xi_0).$$

The class of functions A will be the same for ϕ_1 and for ϕ_2.

—— · ——

FIO:



Undeciphered passage. Two entries by X, Y are heavily crossed out.

Original source details: GELFAND-P000591

X, Y — 2 manifolds, Λ — a closed conic Lagr. manifold without singularities in $T^*(X \times Y) \setminus 0$.

Then locally Λ is given by means of a phase function $\phi(x,y,\theta)$. (In a neighbourhood of the point (x_0,ξ_0,y_0,η_0))



Undeciphered passage. A mark before “point” is heavily crossed out.

Original source details: GELFAND-P000591

and choose an amplitude a and define the operator by means of the kernel

$$\int a(x,y,\theta)e^{i\phi(x,y,\theta)}u(x,y)\,dx\,dy\,d\theta$$

Undeciphered passage. A short entry to the right of the integral is crossed out.



Original source details: GELFAND-P000591

GELFAND-P000592

[source number: 4]

Kirillov. Fact: M — a sympl. manifold, ω — the corresponding form.

Undeciphered passage. A short entry after "form" is crossed out.



Original source details: GELFAND-P000592

on M there is a Lagrangian foliation.

Diagram and typed description.



Original source details: GELFAND-P000592

Then there is an invariantly defined affine structure on its leaves.

Uncertain reading. A mark before "If" is crossed out.

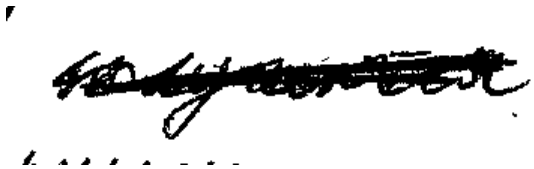


Original source details: GELFAND-P000592

If the functions on M are constant on the leaves of the foliation, then their gradients (vectors corresponding to the covectors $d\varphi$) define parallel transport there. The corresponding vector fields commute.

$$\varphi \longrightarrow d\varphi \longrightarrow \xi_\varphi.$$

— these vector fields ~~commute~~ are called parallel.



Uncertain reading. The crossed-out word tentatively repeats “commute”.

Original source details: GELFAND-P000592

Corollary: the leaves — $\mathbb{R}^n$ or tori.

“Kokas.” construction — an example of a Lagr. foliation.



Uncertain reading. The overwritten abbreviation tentatively reads “Кокас.” (“Kokas.”).

Original source details: GELFAND-P000592

— · —

Tulovskii.

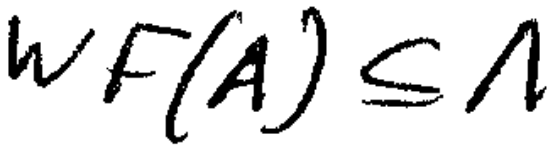
In the general case:

$$\sum_j \int a_j(x,y,\theta_j) e^{i\phi_j(x,y,\theta_j)} u(x,y)\, dx\, dy\, d\theta_j$$

— this is an FIO.

~~Theorem.~~

This is a class of operators for which $\mathrm{WF}(A) \subseteq \Lambda$ and which act well on rapidly oscillating exponentials.



Uncertain reading. The sign between $\mathrm{WF}(A)$ and Λ reads as $\subseteq$.

Original source details: GELFAND-P000592



Uncertain reading. The overwritten word reads as “which”.

Original source details: GELFAND-P000592

Let (x, Ψ'_x) be transverse to Λ.

$$A(e^{it\Psi}\varphi) \sim e^{it\phi} \sum_{j=0}^{\infty} b_j(x) t^{-\lambda_j}$$

Then this is an FIO.

— · —

GELFAND-P000593

[The upper part of the frame contains no text.]

10/III–75. Gelfand's sem.

Tulovskii. (continued) $\Omega_x \subseteq \mathbb{R}^{n_x}$, $\Omega_y \subseteq \mathbb{R}^{n_y}$.

$$Au(x) = \int a(x,y,\theta) e^{i\phi(x,y,\theta)} u(y)\, dy\, d\theta$$

$$A : \mathcal{E}'(\Omega_y) \longrightarrow \mathcal{D}'(\Omega_x)$$

With Φ is associated a Lagr. manifold

$$\Lambda \subset (\Omega_x \times \Omega_y) \times (\mathbb{R}^{n_x+n_y} \setminus 0)$$



Original source details: GELFAND-P000593

Uncertain reading. The overwritten sign after Λ reads as $\subset$.

PDO case: $\Lambda = (x, \xi, x, -\xi)$.



Original source details: GELFAND-P000593

Undeciphered passage. A short entry to the right of the formula is crossed out.

Question from G.: it would be good to construct this in the p-adic case.

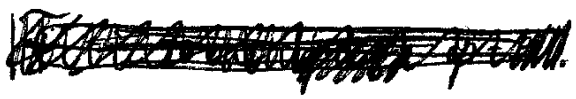
When is the kernel $A(x,y)$ — the kernel of an FIO?

Tulovskii's answer:

$$1) \quad \mathrm{WF}(A) \subset \Lambda \qquad (\Lambda \text{ — some Lagr. manifold})$$

$$2) \quad \forall \Psi : \{(x, \Psi'_x, y, \Psi'_y)\} \text{ transverse to } \Lambda \quad \Rightarrow$$

$$\langle A, e^{it\Psi}\varphi(x,y)\rangle \sim \sum e^{it\tilde{\Psi}} a_j(t, \ldots)$$



Original source details: GELFAND-P000593

Undeciphered passage. A line beneath the asymptotic expression is heavily crossed out.

and $\widetilde{\Psi}$ is related to Ψ through Λ.

Thm. 1. $P(x, D_x)$ — a first-order PDO with real principal part $p(x, \xi)$, $p(x_0, \xi_0) = 0$,

$$p'_{x,\xi}(x_0, \xi_0) \neq 0 \quad \Rightarrow \quad \exists\, 2 \text{ FIOs}$$

$$A : \mathcal{E}'(\Omega_x) \longrightarrow \mathcal{E}'(\mathbb{R}^n),$$
$$B : \mathcal{E}'(\mathbb{R}^n) \longrightarrow \mathcal{E}'(\Omega_x)$$

$$\mathrm{WF}\left[P - B\frac{\partial}{\partial X_n}A\right] \not\ni (x_0, \xi_0, x_0, -\xi_0)$$



Uncertain reading. There are crossed-out strokes immediately after the first ξ_0.

Original source details: GELFAND-P000593

Moreover

$$\mathrm{WF}(B \circ A - I) \not\ni (x_0, \xi_0; x_0, -\xi_0)$$



Uncertain reading. The overwritten letters read as B and A. A small circle is visible between them.

Original source details: GELFAND-P000593

~~Theorem~~

Thm. 2 $P(x, D_x)u = f$,

$$H_p = \sum \left(\frac{\partial p}{\partial x_j}\frac{\partial}{\partial \xi_j} - \frac{\partial p}{\partial \xi_j}\frac{\partial}{\partial x_j}\right)$$



Undeciphered passage. A short mark between f and H_p is crossed out.

Original source details: GELFAND-P000593

$$\mathrm{WF}(u) \setminus \mathrm{WF}(f) \subset \{p(x, \xi) = 0\}$$

and invar. under the Ham. field H_p.

next time R. L. Dobrushin.

GELFAND-P000594

I. Bernstein. WF of a left ideal of diff. operators with polynomial coeffs.



Uncertain reading. The crossed-out abbreviation after “ideal” tentatively reads “oper.”.

Original source details: GELFAND-P000594

$$= \{(x_0, \xi_0) : \dddot{-} p(x_0, \xi_0) = 0, \ \forall p\}$$



Uncertain reading. The initial sign before p is heavily overwritten. The ending $\forall p$ is clear.

Original source details: GELFAND-P000594

Thm. $\dim \mathrm{WF} \geq \dim X$.

GELFAND-P000595

[The upper part of the frame contains no text.]

17/III–75. Gelfand's sem.

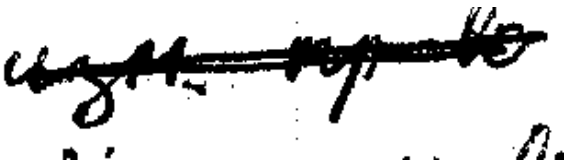
Seminar programme:

Dobrushin R. L.

Objects

1) random fields with a discrete argument

X — ~~measurable space~~ a finite set



Uncertain reading. The crossed-out expression tentatively reads "measurable space".

Original source details: GELFAND-P000595

T — a countable set

$$X^T = (x_t, \ t \in T)$$

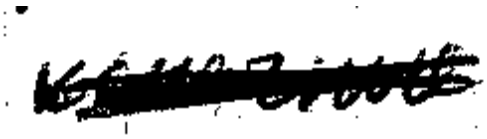
A random field — a prob. measure on X^T.

Physical interp. — spin systems

2) point random fields

T — a continuous space $\mathbb{R}^\nu$

Realizations — ~~finite~~ subsets of $\mathbb{R}^\nu$, $\cap$ of which with $\forall$ finite volume is finite.



Uncertain reading. The crossed-out word tentatively reads "finite".

Original source details: GELFAND-P000595

A measure on realizations — a point random field.

$$3) \qquad \underset{\substack{\downarrow \\ \text{functions (real)}}}{X(t)} \quad , \quad t \in \mathbb{R}^\nu \qquad S^*(\mathbb{R}^\nu)$$

a random field — a measure on S^*

$\Rightarrow$ quantum field theory.

GELFAND-P000596

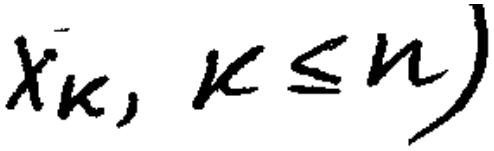
Problem: description of random fields

$$T = \mathbb{Z}, \qquad X^{\mathbb{Z}}\ (x_t,\ t \in \mathbb{Z})$$

Def

$$\Pr(x_n \mid x_k,\ k \le n) \underset{\text{Markov}}{=} \Pr(x_n \mid x_{n-1}) \qquad \text{(Markov chain)}$$

Cond. prob. is known.



Uncertain reading. The condition on the left reads $k \le n$.

Original source details: GELFAND-P000596

Task: first specify the conditional probability, then see which measures have these conditional probabilities.

—— · ——

$$V \subset \mathbb{Z}^\nu, \quad \text{specify} \quad \Pr\{\xi_n,\ n \in V \mid \xi_k,\ k \in \mathbb{Z}^\nu \setminus V\}.$$

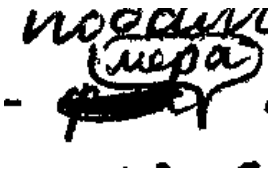
—— · ——

Conditional probability: $(X, \mathcal{B}, \mu)$ — a prob. space.

$$\mu(A \mid B) = \frac{\mu(A \cap B)}{\mu(B)} \qquad \text{— insufficient}$$

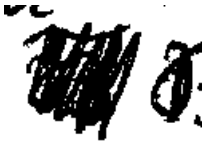
Let $\widetilde{\mathcal{B}}$ — be a subalgebra in $\mathcal{B}$

Conditional probability — measure on $\mathcal{B}$, depending on $x \in X$



Uncertain reading. The crossed-out word under the insertion “measure” tentatively reads “function”.

Original source details: GELFAND-P000596



Uncertain reading. An entry before $\mathcal{B}$ is crossed out.

Original source details: GELFAND-P000596

$$\mu(\cdot \mid x), \quad x \in X. \qquad\qquad \mu(B \mid x), \quad B \in \mathcal{B},\ x \in X.$$

1) $\forall B:\ \mu(B \mid x)$ is measurable on $\widetilde{\mathcal{B}}$

2)

$$\int \psi(x)\varphi(x)\mu(dx) = \int \psi(x)\mu(dx \mid y)\varphi(y)\mu(dy)$$

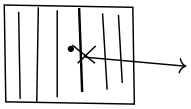
(φ is measurable with respect to $\widetilde{\mathcal{B}}$)

Then the corresponding μ is called the conditional probability distribution.

In a broad class of cases $\exists$ and uniqueness..



Diagram and typed description.



The exact shape of the crossing marks on the middle segment is unclear.

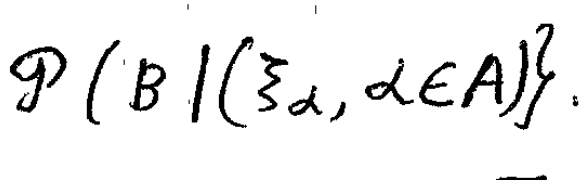
Original source details: GELFAND-P000596

here the measure $\overset{\mu(B|x)}{\uparrow}$ sits on the segment. One must take Borel sets (rather than Lebesgue-measurable ones).

In this example:

$$\int \psi(x_1, x_2)\varphi(x_2)\,dx_1\,dx_2 = \int \left(\int \psi(x_1, x_2)\,dx_1 \right) \varphi(x_2)\,dx_2$$

$$\mathcal{P}\big(B \mid (\xi_\alpha, \alpha \in A)\big\}. \qquad \mathcal{P}_{\mathcal{B}(\xi_\alpha, \alpha \in A)}(B \mid X)$$



Uncertain reading. The first expression uses round parentheses and ends with a closing brace.

Original source details: GELFAND-P000596

$$\xi_\alpha : X \underset{\text{functions}}{\longrightarrow} F_\alpha \qquad \text{or one} \quad \xi : X \to F \quad \text{(into the direct product)}$$

points of F — layers

preimages of sets in F — $\widetilde{\mathcal{B}}$

↓ there the σ-algebra $\mathcal{B}_F$.

—— · ——

GELFAND-P000597

Gelfand Sem. 31/III-75

Polyakov (Sasha) On the idea of [inserted above the line and outlined: “spontaneously”] broken symmetries

“Bubbles in the ether of quant. field theory are particles”

Skyrme; Nielsen, Olsen, Faddeev, Polyakov, ’t Hooft.

Next time — Nevanlinna theory (following Griffiths)

GELFAND-P000598

7/IV-75. Gelfand Sem.

King (?) Nevanlinna theory following Griffiths.



Uncertain reading. The manuscript places a question mark in parentheses after “King”.

Original source details: GELFAND-P000598

2 ideas. 1) hyperbolic analysis

$$\underset{\text{holomorphic}}{f} : X \to V \qquad \text{negatively curved}$$

$\Rightarrow$ there are some restr. on behaviour of f.

2) “counting” of volume of divisers of analytic sets by residue calculus.

First 2)

g — holom. function, $D = \{g^{-1}(0)\}$



Diagram and typed description.



In the left margin two curved strokes cross a nearly vertical line. Their direction and exact closure are unclear.

Original source details: GELFAND-P000598

integral over a neighbourhood of a form (of lower order).

Let η — any form of deg. r on $\mathbb{C}^n \setminus D$

[outlined above the line] $\leftarrow$ this too is a form

$$\eta = \frac{dg}{2\pi i g} \wedge \zeta + \xi, \qquad \zeta, \xi \text{ — forms, [crossed out] then}$$

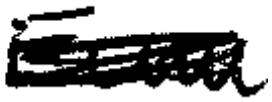


Uncertain reading. An entry before “then” is crossed out.

Original source details: GELFAND-P000598

$$\lim_{\varepsilon\to 0}\int_{C(\varepsilon)}\eta = \zeta|_D$$

η, ζ — currents (current) (functionals on forms): ~~so~~ the easiest way to define them:



Uncertain reading. A short entry before η, ζ is crossed out.

Original source details: GELFAND-P000598

Current:

$$T : A_c^k(V) \to \mathbb{C}$$

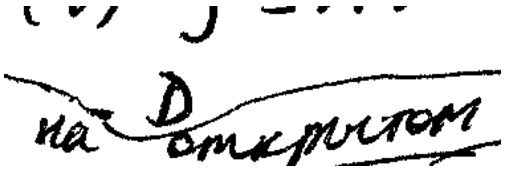
$$\lambda \in L^1_{\text{loc}}\ r\text{-form} \quad \Rightarrow \quad [\lambda](\varphi) = \int_X \lambda\wedge\varphi,$$

$$dT(\varphi) = \pm T(d\varphi).$$

The form $\zeta|_D$ defines a current:

$$[\zeta|_D](\varphi) = \int \zeta\wedge\varphi.$$

Def. Let η — L^1_{loc} form, $\in C^\infty$ on ^{D} an open dense set

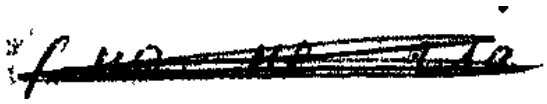


Uncertain reading. D is written above the line, next to the outline.

Original source details: GELFAND-P000598

$$\operatorname{Res}\eta = [\zeta|_D] := d[\eta] - [d\eta]$$

[at right below the outlined line] η — an ordinary form



Uncertain reading. An entry below “ordinary form” is crossed out. It tentatively begins “(on”.

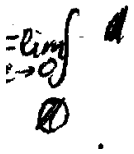
Original source details: GELFAND-P000598

$$d[\eta](\varphi) \underset{\text{def}}{=} [\eta](d\varphi) = \int_V \eta \wedge d\varphi$$
$$= \lim_{\varepsilon \to 0} \int_{V - U_\varepsilon(D)} \eta \wedge d\varphi =$$
$$= \int_V d\eta \wedge \varphi - \lim_{\varepsilon \to 0} \int_{\partial U_\varepsilon(D)}$$
$$= \int_V d\eta \wedge \varphi - \int_D \dots$$

Example:

$$\eta = \frac{dz}{z}, \qquad [\eta](\varphi) = \int \frac{dz \wedge \varphi}{z}$$

$$d[\eta](\varphi) - [\eta](d\varphi) = \lim_{\varepsilon \to 0} \int_{\text{[overwritten]}} \varphi \, dz = \varphi(0).$$

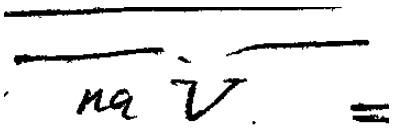


Uncertain reading. The round mark below the integral is overwritten and may be O. A short stroke above it is also crossed out.

Original source details: GELFAND-P000598

Res η — a current.!

g — holom. function on $\bar{V}$.

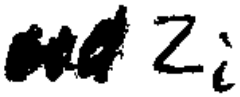


Uncertain reading. There are separate strokes above V, tentatively read as an overbar.

Original source details: GELFAND-P000598

$$\Rightarrow \quad \operatorname{Res}\left[\frac{dg}{2\pi i g}\right] = [Z],$$

$$Z = \{g = 0\}, \quad Z = \bigcup Z_i, \quad \operatorname{ord} g = m_i \text{ [overwritten] } Z_i$$



Uncertain reading. The word before Z_i is overwritten and may read “on”.

Original source details: GELFAND-P000598

$$Z = \sum m_i \int_{Z_i} \qquad \text{(as a current)}$$

GELFAND-P000599

Let $X \subset V$; Codim $X = 1$.

$$X \cap U_i = \{g_i = 0\} \qquad f_{ij} = \frac{g_i}{g_j} \text{ in } U_i \cap U_j$$



Diagram and typed description.

$$\begin{array}{c} L_X \\ \downarrow \\ V \end{array}$$

Original source details: GELFAND-P000599

$$\sigma = \{g_i\}, \quad \sigma \in \Gamma(L_X).$$

$\sigma = 0$ on X.

—— · ——

Let L have a Herm. metric (L — a 1-dim. bundle)
We can consider $|\sigma|^2 \in C^\infty$ on V and $= 0$ on X.

$$\eta = \frac{1}{2\pi i}\partial \log |\sigma|^2, \qquad \partial \log |z|^2 = \frac{dz}{z}.$$

Then Res $\eta = [X]$.

$$[d\eta] = \frac{\bar\partial\partial}{2\pi i} \log |\sigma|^2 = \frac{\bar\partial\partial}{2\pi i} \log h_i = c(L_D).$$

(Chern class)

where $|\sigma|^2 = h_i |g_i|^2$

$$\partial\bar\partial \log \bar g_i = 0$$



Uncertain reading. The separate line has a bar over the second ∂ and over g_i.

Original source details: GELFAND-P000599

De Rham's thm. holds! (in currents):

$$X = d[\eta] + [c(L_D)]$$

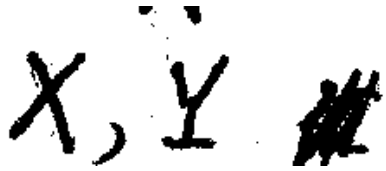
(the Chern class is cohomologous to the zero set of any section)!

—— · ——

Let ψ — a closed form. [inserted above the line and outlined: Then]

$$\int_X \psi = \int_V c(L_D) \wedge \psi.$$

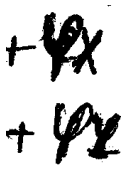
Bézout-type theorems. Let X, Y [crossed out] — two curves in $\mathbb{CP}^2$.



Original source details: GELFAND-P000599

Uncertain reading. A short mark after X, Y is crossed out.

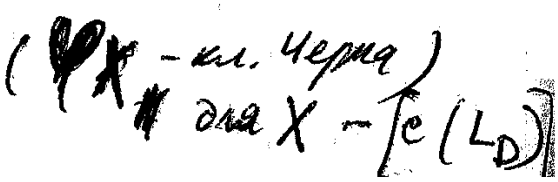
$$X = d[\eta_X] + \varphi_X$$
$$Y = d[\eta_Y] + \varphi_Y$$



Original source details: GELFAND-P000599

Uncertain reading. The last term of each equation is heavily inked and reads φ_X, φ_Y.

(φ_X — Chern cl.)
[crossed out] for X — $[c(L_D)]$



Original source details: GELFAND-P000599

Uncertain reading. A small mark before "for" is crossed out.

Consider

$$\int_V \varphi_X \wedge \varphi_Y = \int_X \varphi_Y = [X \wedge Y](1)$$

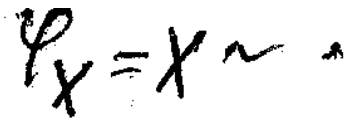
(number of intersection points =
= prod. of Chern classes)

Example: $V = \mathbb{P}^2, \quad X, Y = H$. (hyperpl.)

$$d_X = \deg X = \#(X \cap H) = \int_X \omega$$

(ω — std. form dual to the hyperpl. section)

$$\varphi_X = X \sim \text{[small mark]}$$



Uncertain reading. The short mark after $\sim$ resembles a small angle.

Original source details: GELFAND-P000599

$$\int_{\mathbb{P}^2} \varphi_X \wedge \varphi_Y = d_X d_Y \int_{\mathbb{P}^2} \omega^2 = d_X \cdot d_Y$$

— this is the proof of Bézout's thm. on the number of intersection points of curves.

GELFAND-P000600

X with singularities $\Rightarrow$ the thm. still holds (Lelong, 1953)

—— · ——

1) idea.

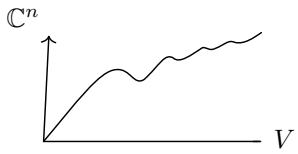
$$f : \mathbb{C}^n \xrightarrow[\text{holom.}]{} V$$

[below $\mathbb{C}^n$:] $\varphi \leftarrow$ Eucl. Kähler
[arrow to V:] compact. Kaehler ([inserted above the line and outlined: with] form ω).
Let $B(r)$ — be the ball of rad. r.

$$\text{vol}(f(B(r))) = \int_{B(r)} f^*(\omega^n)$$



Diagram and typed description.



The right end of the horizontal axis is thickened. An arrowhead cannot be established.

Original source details: GELFAND-P000600

[at right] Γ [overwritten subscript] [inserted above the line and outlined: part] of the graph of f. over $B(r)$



Uncertain reading. The subscript of Γ is heavily overwritten and may be f or r.

Original source details: GELFAND-P000600

$$\int_{\Gamma(r)} (\varphi+\omega)^n = \sum \binom{n}{k} \int_{B(r)} \varphi^k \wedge f^*\omega^{n-k}$$

Def.

$$T_k(f,r) \underset{\text{def}}{=} \int_{B(r)} \varphi^{n-k} \wedge f^*(\omega^k)$$

In many cases $T_k = O(T_1)$ (growth in r).

[below the line] of the same order.

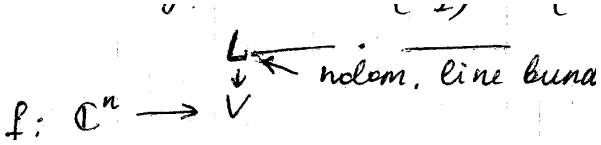


Diagram and typed description.

$L \longleftarrow$ holom. line bundle.

$\downarrow$

$f : \mathbb{C}^n \longrightarrow V$

Original source details: GELFAND-P000600

Let $0 < \omega = c(L)$ — its Chern class

Let $\sigma \in O(L), \quad D = \{\sigma = 0\}$.

$$d[\eta] - [\omega] = [D]; \qquad D_f = f^{-1}D.$$

$$d[f^*\eta] - [f^*\omega] = [f^{-1}D] \quad \text{(on } \mathbb{C}^n\text{)}.$$

Mult. by $\chi_{B(r)}\varphi^{n-1}$:

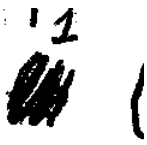
$$\underbrace{\int_{S(r)} \varphi^{n-1} \wedge f^*\omega}_{\text{“}T_1} = \int_{D_f} \chi_{B(r)}\varphi^{n-1}.$$



Uncertain reading. The integral subscript tentatively reads $S(r)$.

Original source details: GELFAND-P000600

$$\eta = \underbrace{\frac{1}{4\pi} \text{[crossed out]} \, (\bar\partial - \partial) \log|\sigma|^2}_{d^c}.$$



Uncertain reading. A mark before $(\bar\partial - \partial)$ is crossed out and may be i.

Original source details: GELFAND-P000600

$$\int_{D_f \cap B(r)} \varphi^{n-1}$$

(volume of the zero set, meas. using the std. φ).

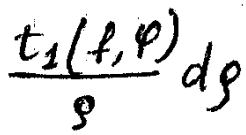


Uncertain reading. The abbreviation after "zero set" tentatively reads "помер." (meas.).

Original source details: GELFAND-P000600

$$T_1 = \int_0^r \frac{t_1(f,\varphi)}{\rho}\, d\rho$$

— $t_1 \leftrightarrow T_1$ in ordinary Nevanlinna th..



Uncertain reading. The argument of t_1 resembles φ. The denominator and differential have ρ.

Original source details: GELFAND-P000600

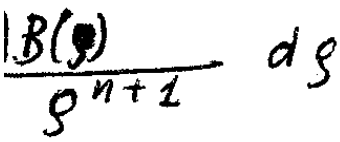
GELFAND-P000601



Uncertain reading. The beginning of a formula $N(\ldots)$ is heavily crossed out.

Original source details: GELFAND-P000601

$$N(D_f, r) = \int_0^r \frac{\int_{D_f \cap B(\rho)} \varphi^{n-1}}{\rho^{n+1}}\, d\rho$$



Uncertain reading. The argument of $B(\rho)$ is overwritten and reads ρ.

Original source details: GELFAND-P000601

$$N(D_f, r) \le T(f, r) + O(1) \qquad \text{(1st Nevanlinna inequality)}$$

GELFAND-P000602

14/IV-75. Gelfand Sem. King (continued).

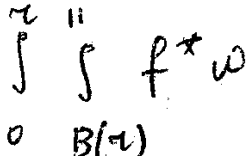
$$\mathbb{C}^n \xrightarrow{f} \underset{\text{projective}}{V^n}, \qquad \underset{\text{divisor.}}{D} \subset V^n, \qquad D \sim c(L_D) = \underset{\text{Kähler form}}{\omega}$$

$$N(D_f, r) \le T(f, r)$$

[below N:] volume of the preimage of the divisor ("number of zeros")
[below T:]

$$\int_0^r \int''_{B(r)} f^*\omega \wedge \psi^{n-1} \frac{dt}{t}$$

("order of growth")



Uncertain reading. Two short vertical strokes are written above the second integral.

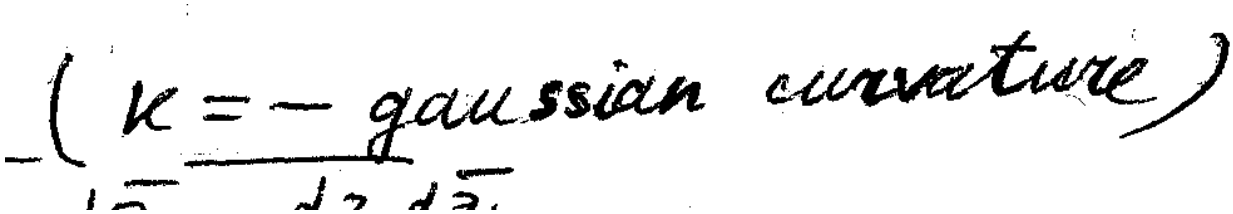
Original source details: GELFAND-P000602

[at right] (1st Nevanlinna thm. — triv.)
uses the residue thm.

$$\operatorname{Ric}\Omega = i\partial\bar\partial \log g \quad \Leftarrow \quad \left(\Omega = \frac{i}{2} g\, dz \wedge d\bar z\right).$$

(Ricci tensor).

$$n = 1 \quad \Rightarrow \quad \operatorname{Ric}\Omega = K\Omega \quad (K = -\text{gaussian curvature})$$



Original source details: GELFAND-P000602

Uncertain reading. The long stroke after $K =$ is read as a minus sign.

Ω — an (n, n) volume form

$$a(\cdot)\, dz_1\, d\bar{z}_1 \ldots dz_n\, d\bar{z}_n$$

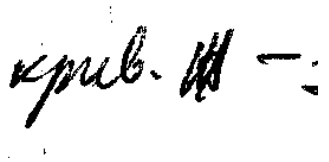
$$\operatorname{Ric}\Omega = \frac{i}{2\pi}\partial\bar{\partial}\log a$$

1) $\operatorname{Ric}\Omega > 0$

2) $(\operatorname{Ric}\Omega)^n \geq \Omega$ (vol. form).

Ahlfors Lemma.

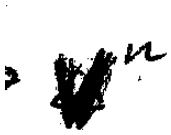
Let B^n or Δ^n have a Bergman form with curv. [overwritten] -1



Uncertain reading. A heavily overwritten mark before -1 resembles H or K.

Original source details: GELFAND-P000602

$\Rightarrow$ $\forall$ holom. $f : B^n \to V^n$; $\quad \Omega_V$ — volume form with neg. curv.



Uncertain reading. The letter before the superscript n is overwritten and tentatively reads V.

Original source details: GELFAND-P000602

$$f^*\Omega_V \leq \Phi_B \text{ — volume form}$$

$\ldots$ ___ $\cdot$ ___ $\ldots$

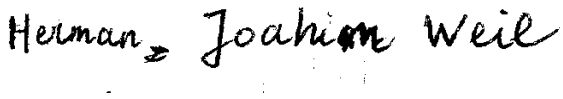
~~Acta~~ Acta Math.; Griffits, King, [crossed out]



Uncertain reading. A small mark after “King,” is crossed out and may read “a”.

Original source details: GELFAND-P000602

Herman $\Rightarrow$ Joahim Weil



Uncertain reading. The name and connecting sign are tentative. The end of “Joahim” is heavily inked.

Original source details: GELFAND-P000602

A nontriv. formulation is needed in the theory of eigenfunctions.

Griffits, Cowen